Emerging signals of a global drift in forest resilience under climate change

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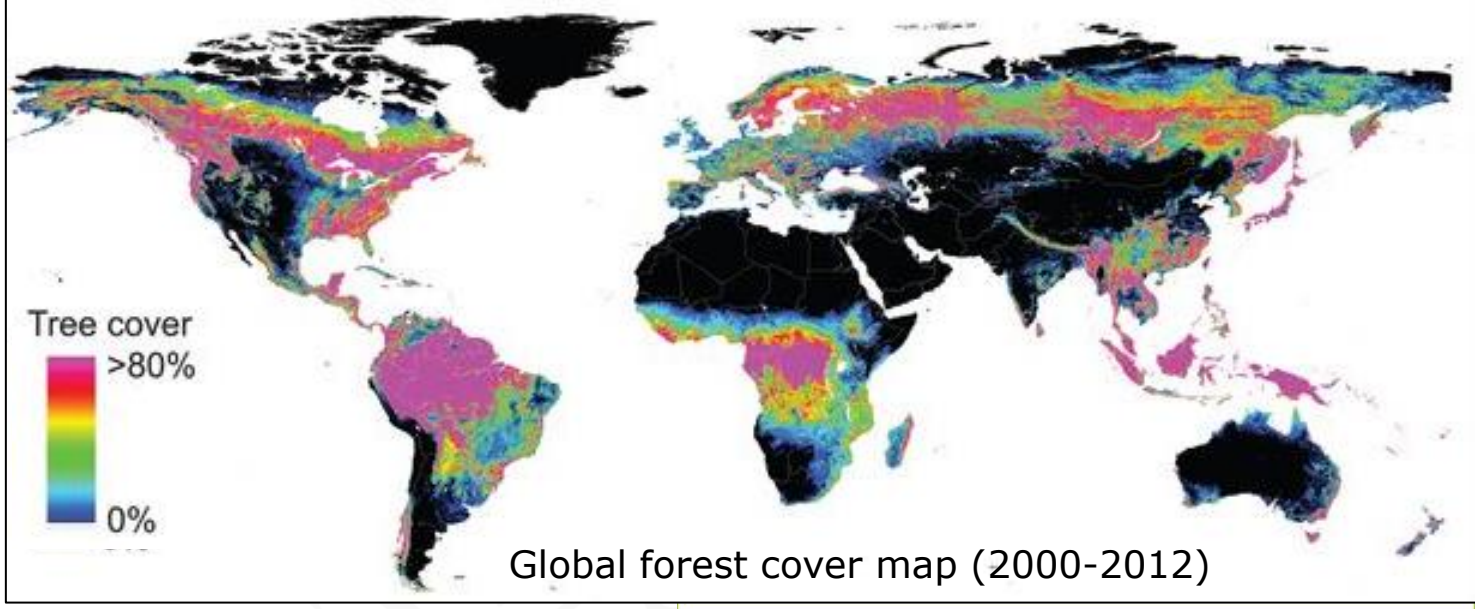
[5] School of Biological Sciences, Washington State University, USA

Rationale

Forests cover about 41 million km^2, about 30% of the land surface

Hansen et al., 2013 Science



Global forest cover map (2000-2012)

Global carbon Budget (2012-2021)

Friedlingstein et al 2022 ESSD

Sources = **Sinks**

35.2 $GtCO_2$/yr
89%

11%
4.5 $GtCO_2$/yr

19.1 $GtCO_2$/yr
48%

29%
11.4 $GtCO_2$/yr

26%
10.5 $GtCO_2$/yr

Goal

The persistence and functionality of these ecosystems are highly dependent on their **resilience, defined as the ability to withstand and recover from environmental perturbations**.

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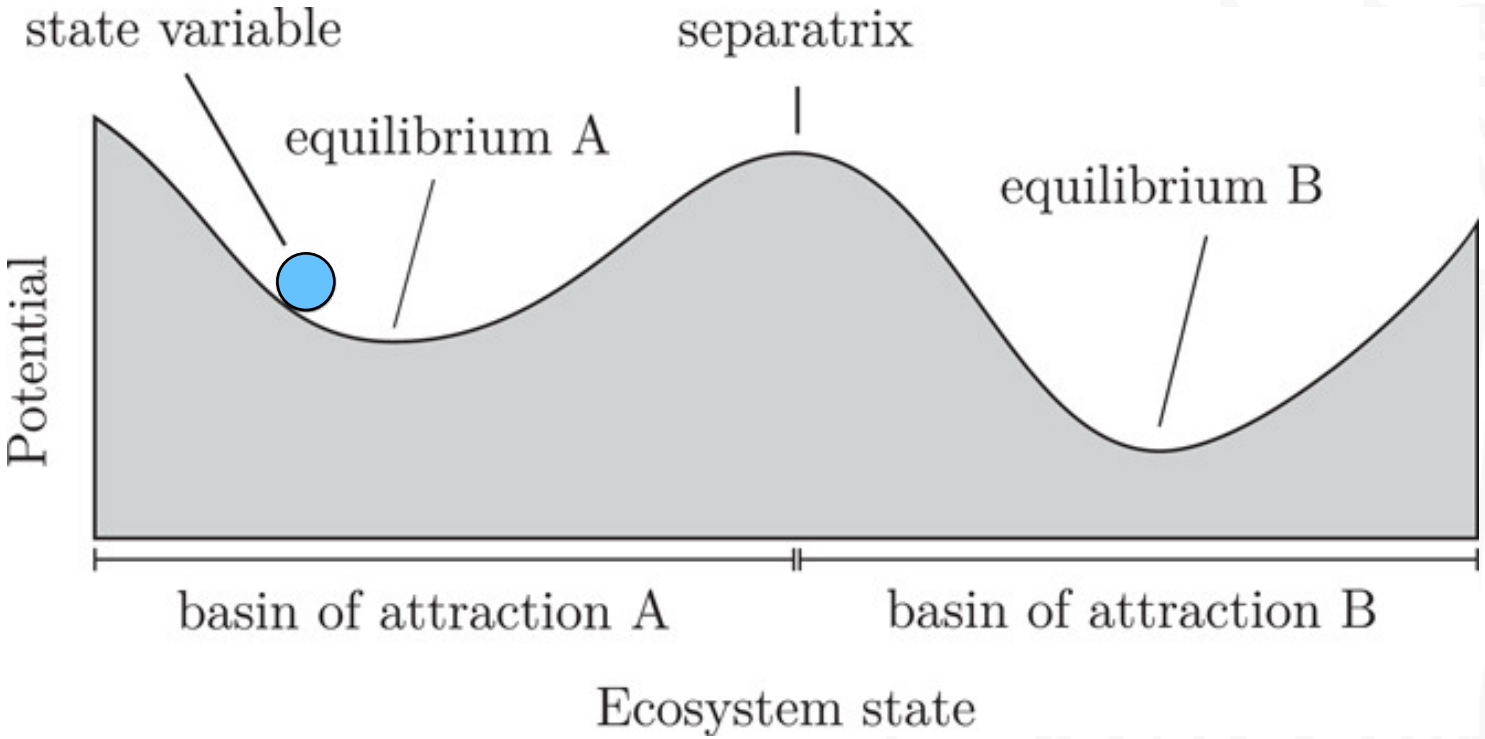
Understanding the underlying mechanisms of forest resilience and its recent dynamics is therefore of paramount importance to develop sound conservation and management plans.

Theoretical framework

Stecher and Baumgärtner 2022



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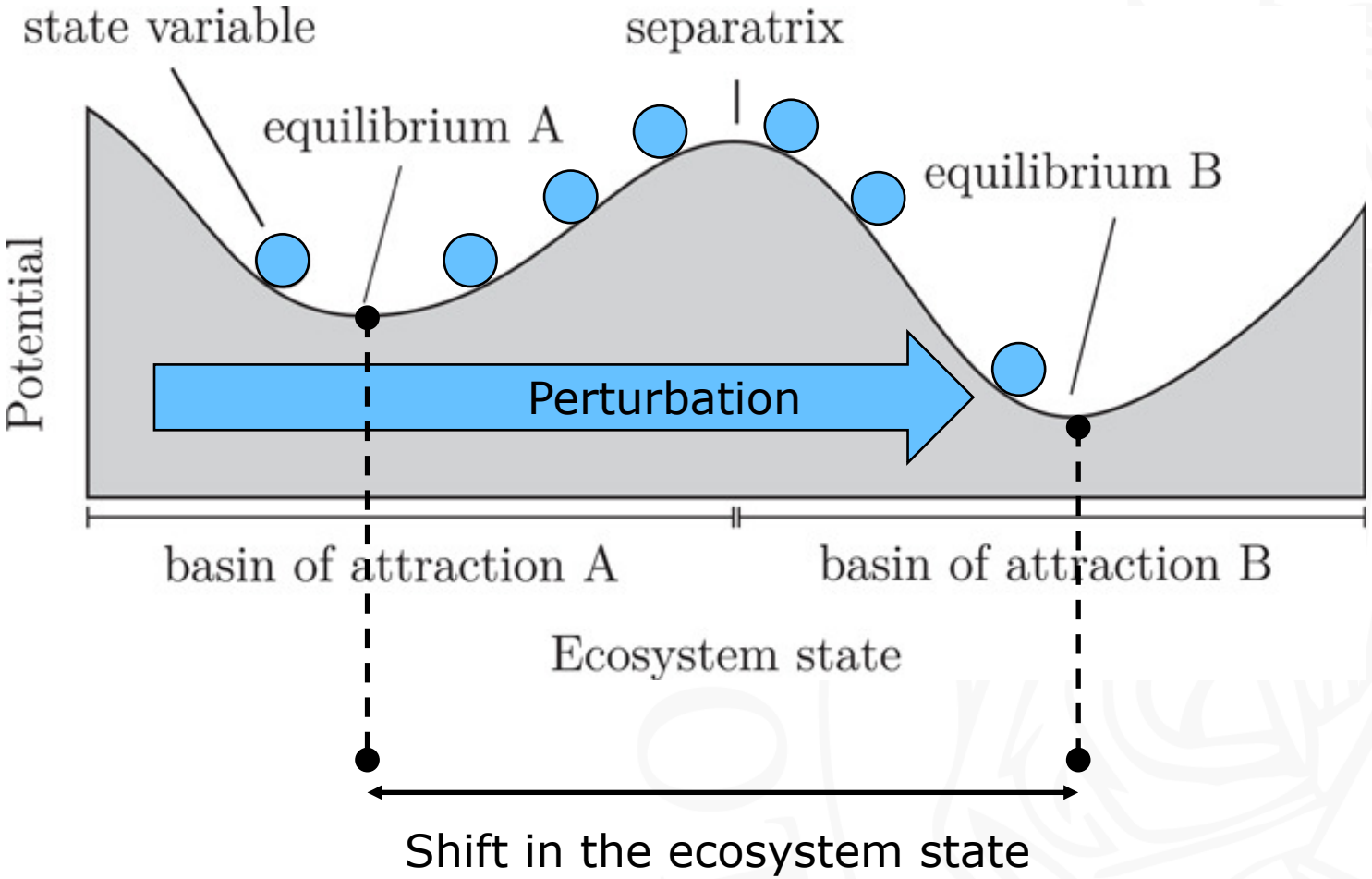

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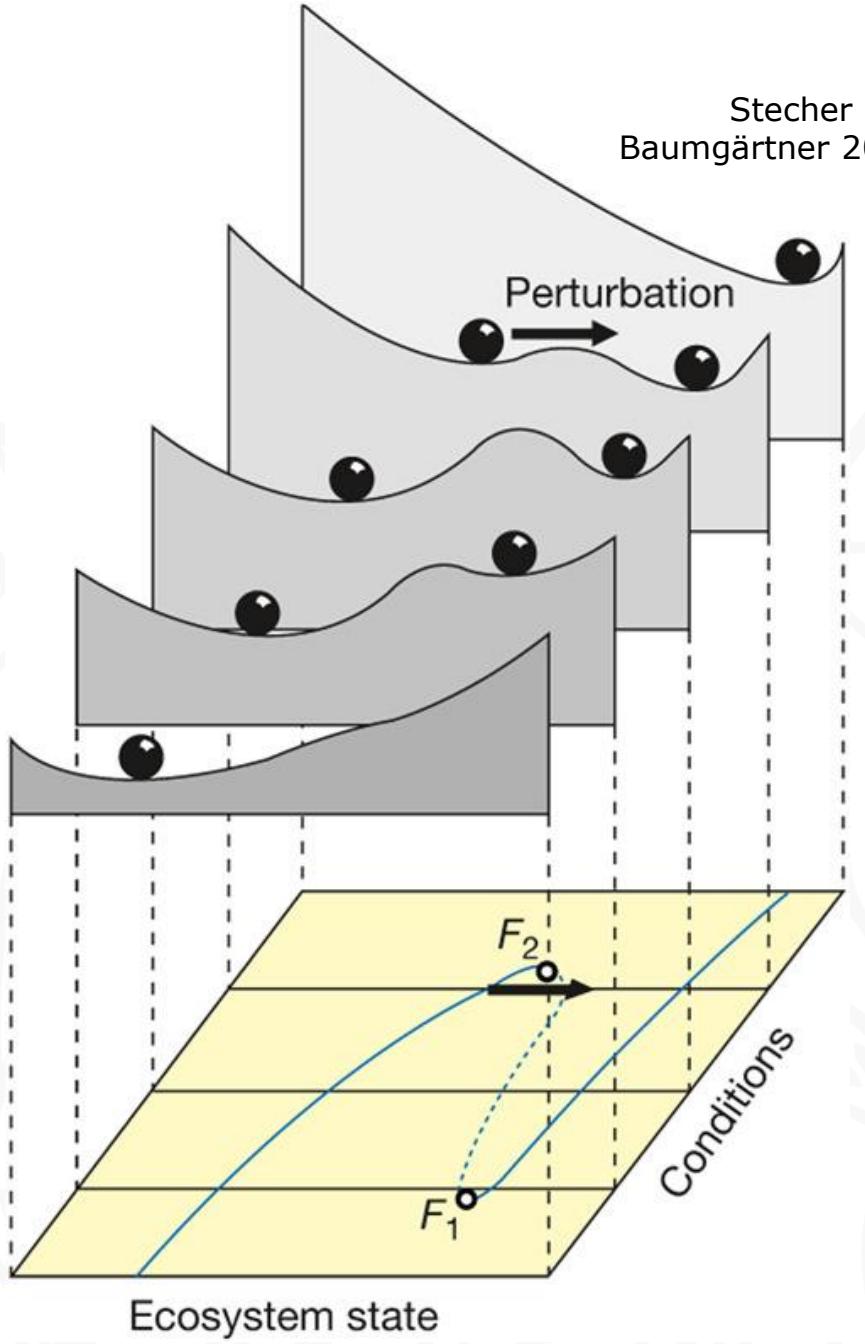

Theoretical framework

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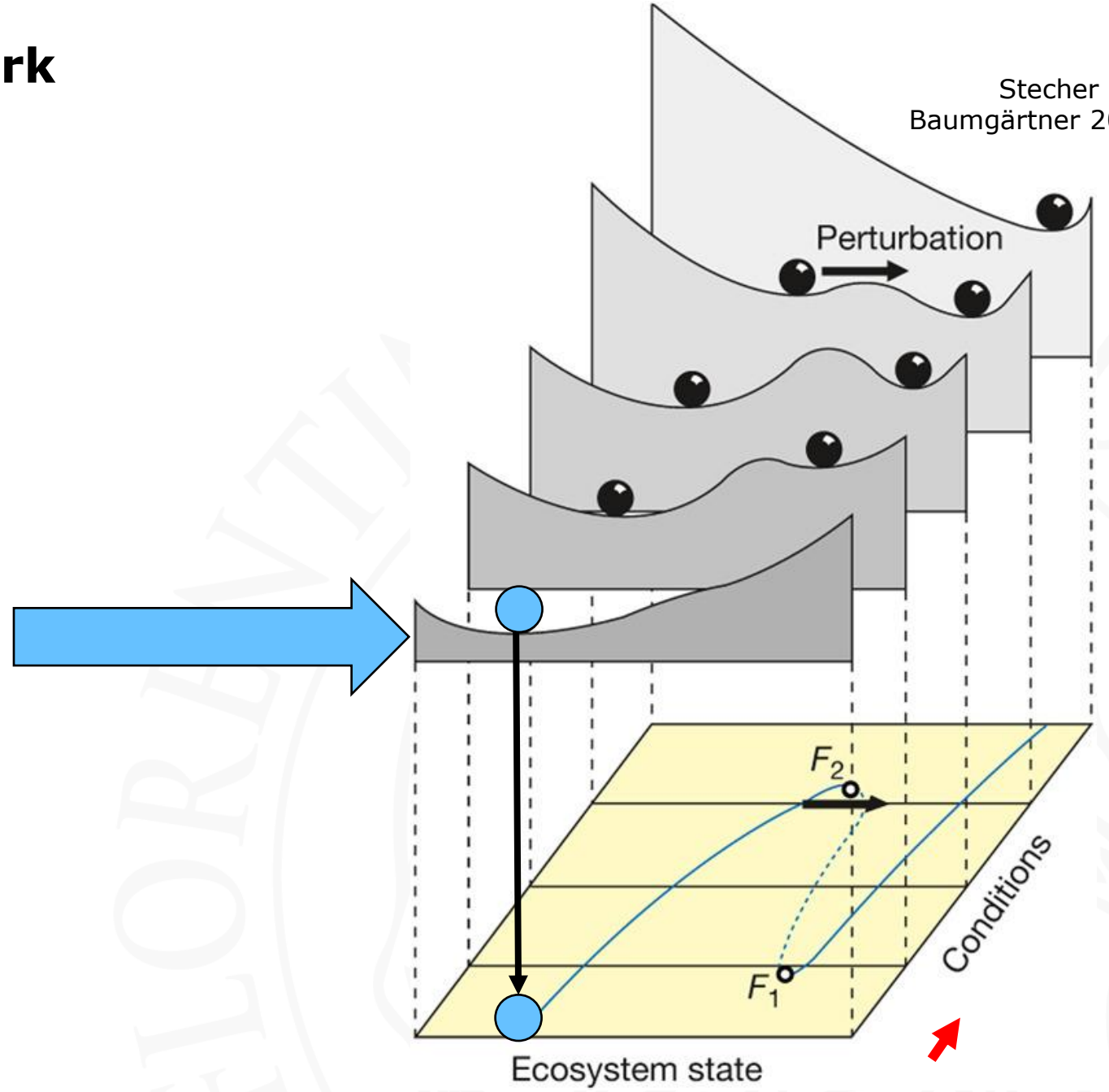
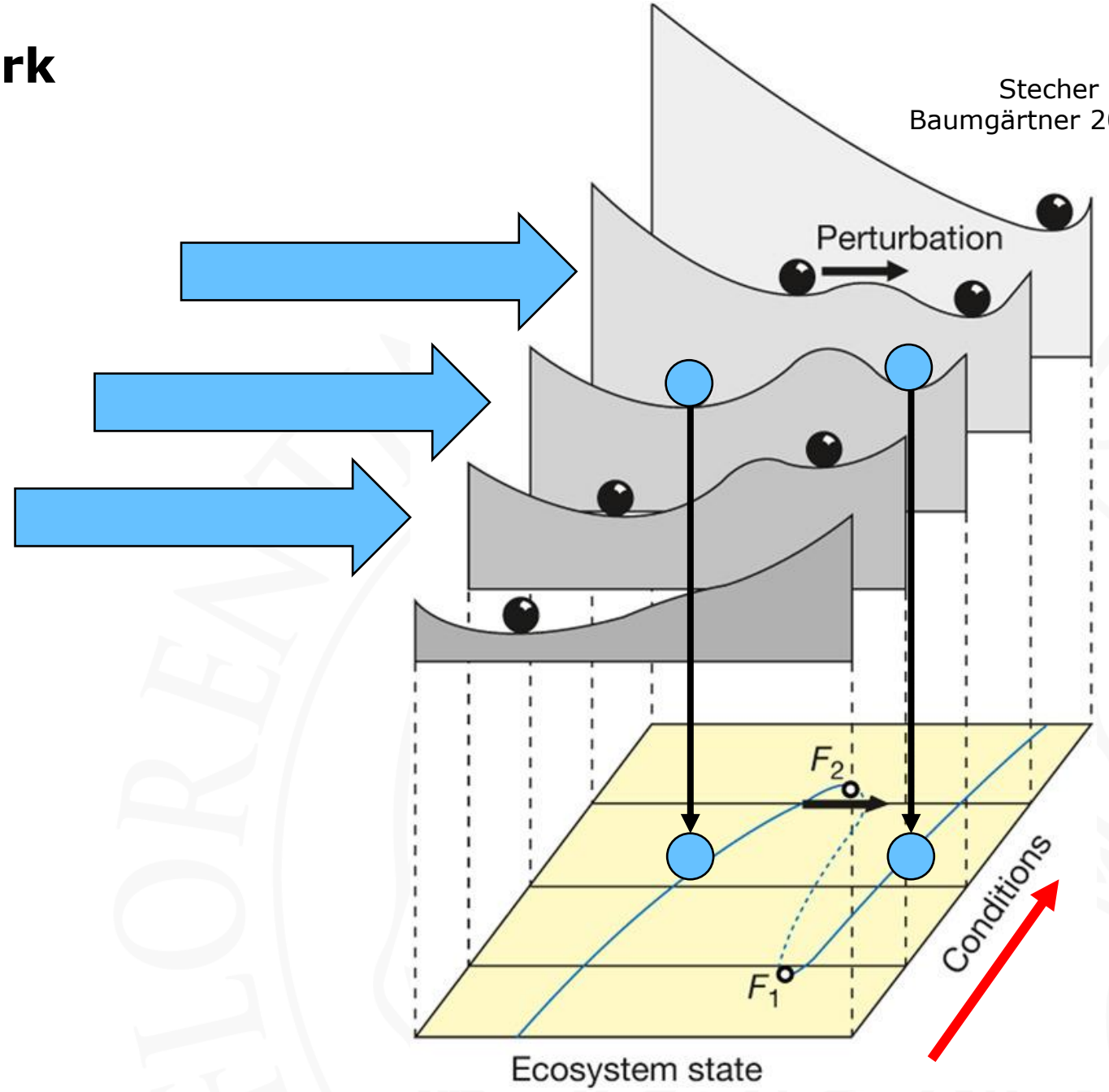
For a low level of conditions only one equilibrium exists at a relatively large value of the state variable



Theoretical framework

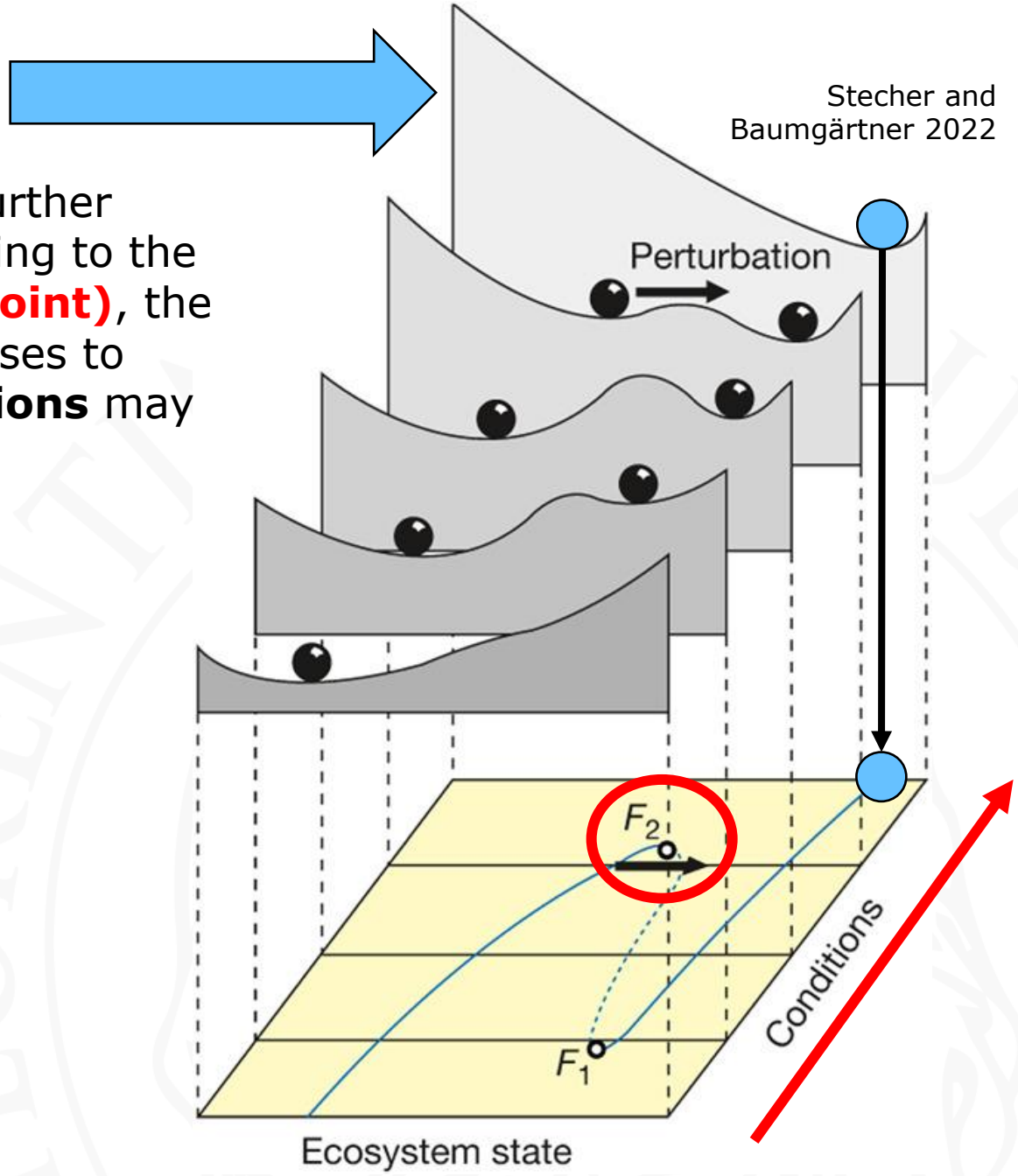
Stecher and Baumgärtner 2022

As conditions increase the stability landscape changes and a second locally stable equilibrium emerges



Theoretical framework

When conditions change further beyond a level corresponding to the **bifurcation F_2 (tipping point)**, the first stable equilibrium ceases to exist, and **critical transitions** may occur



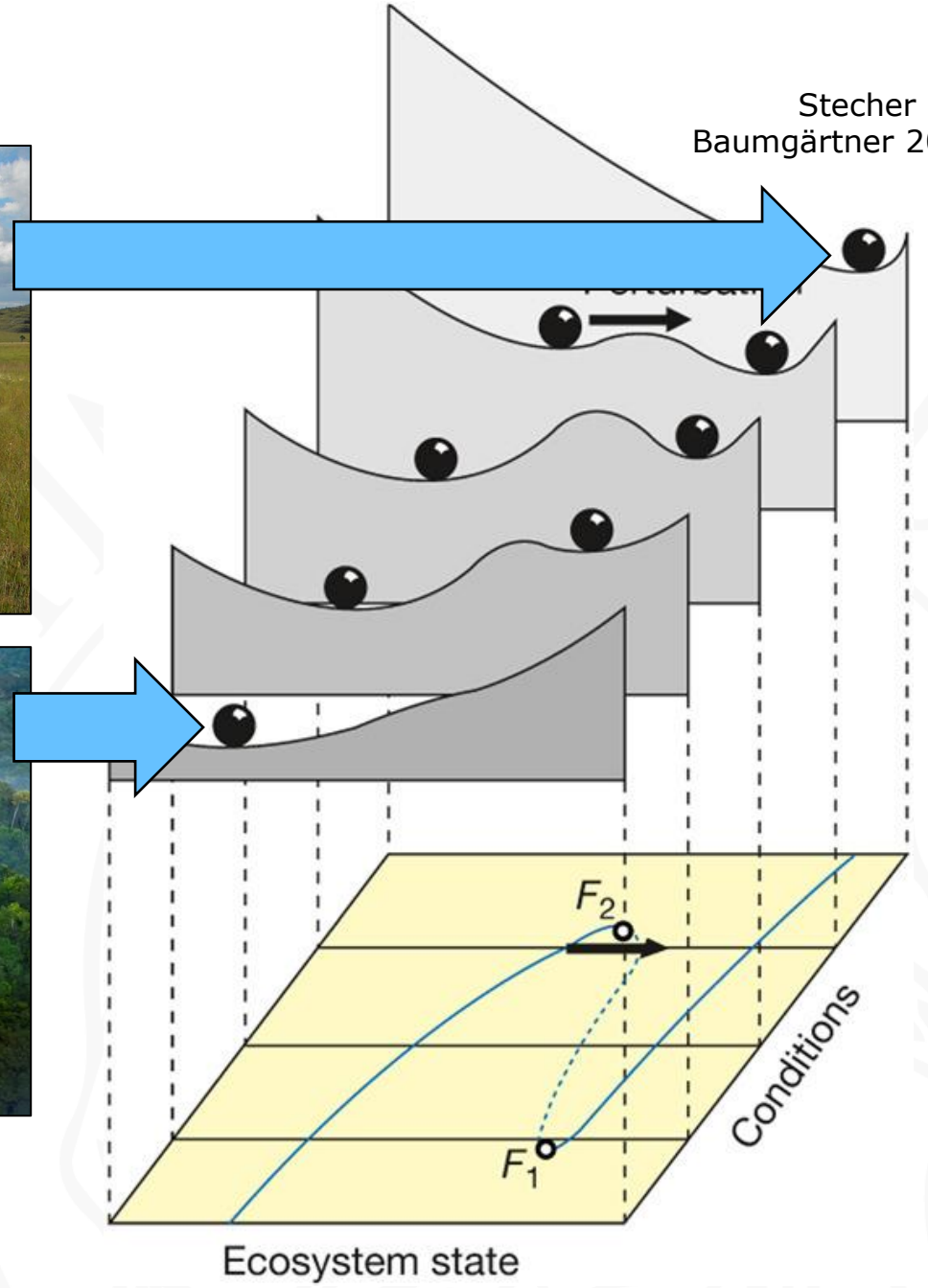
Stecher and Baumgärtner 2022

Theoretical framework

Stecher and Baumgärtner 2022

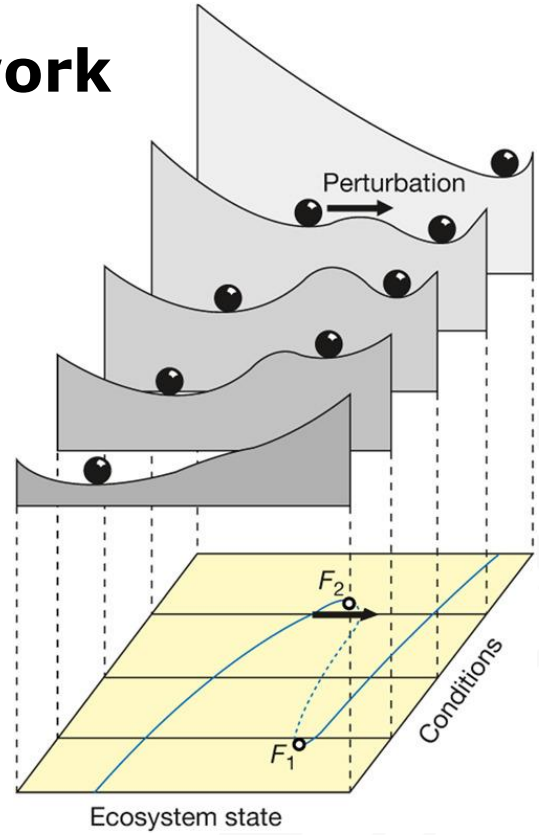
Tropical savanna (Cerrado, Brazil)

Tropical forest (Amazon, Brazil)



Forest and savanna represent alternative stable states, implying critical transitions at tipping points in response to altered climate or other drivers. Hirota et al., 2011 Science

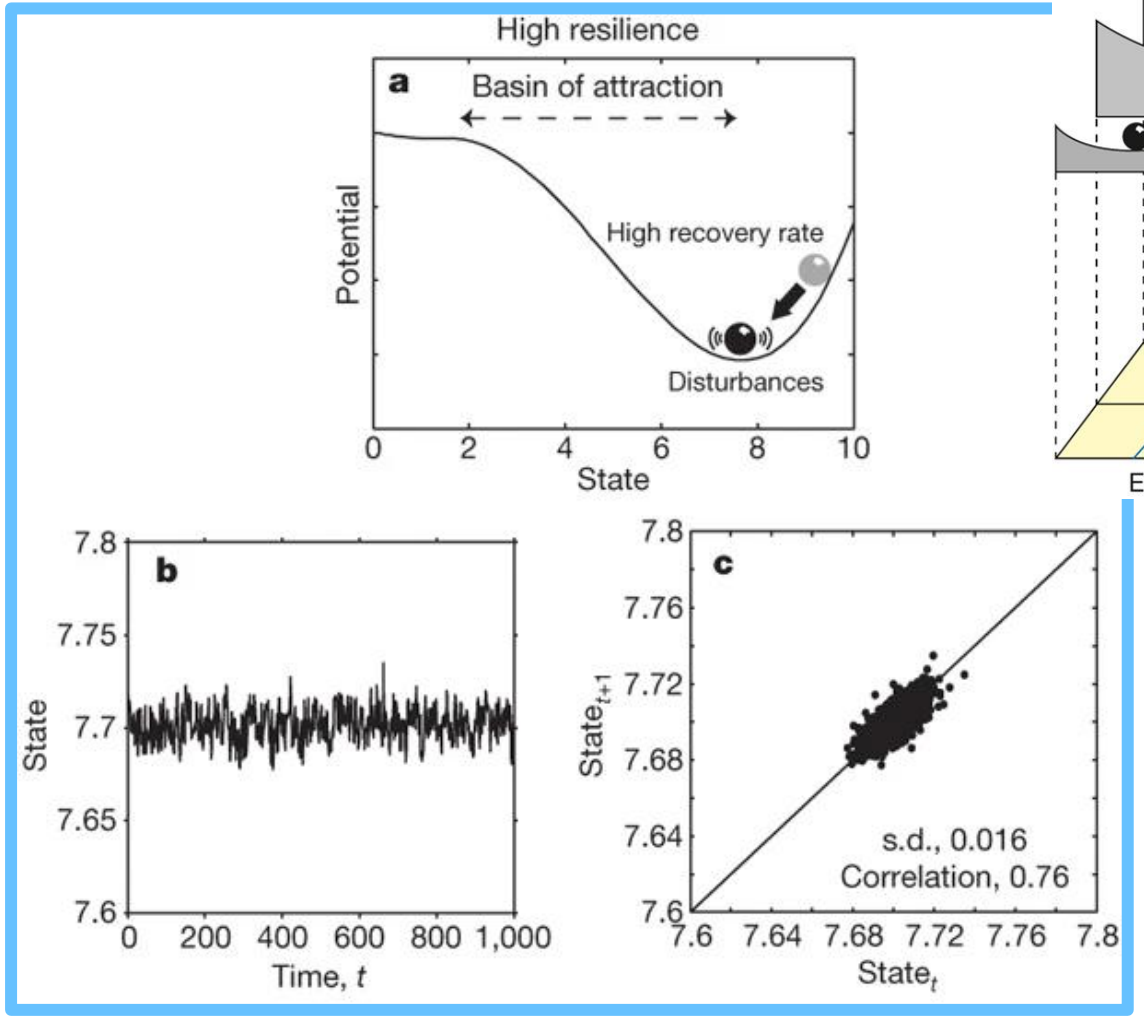
Theoretical framework

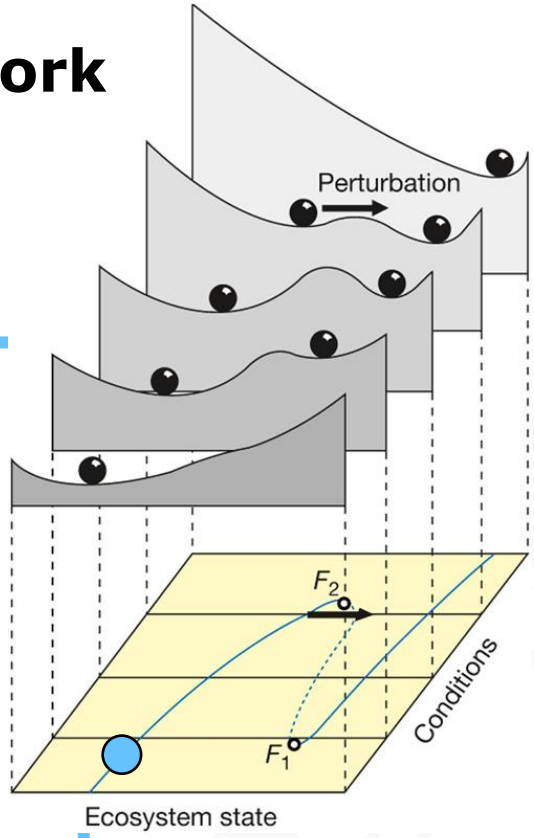


Theoretical framework

Scheffer et al., 2009 Nature

Far from the bifurcation point



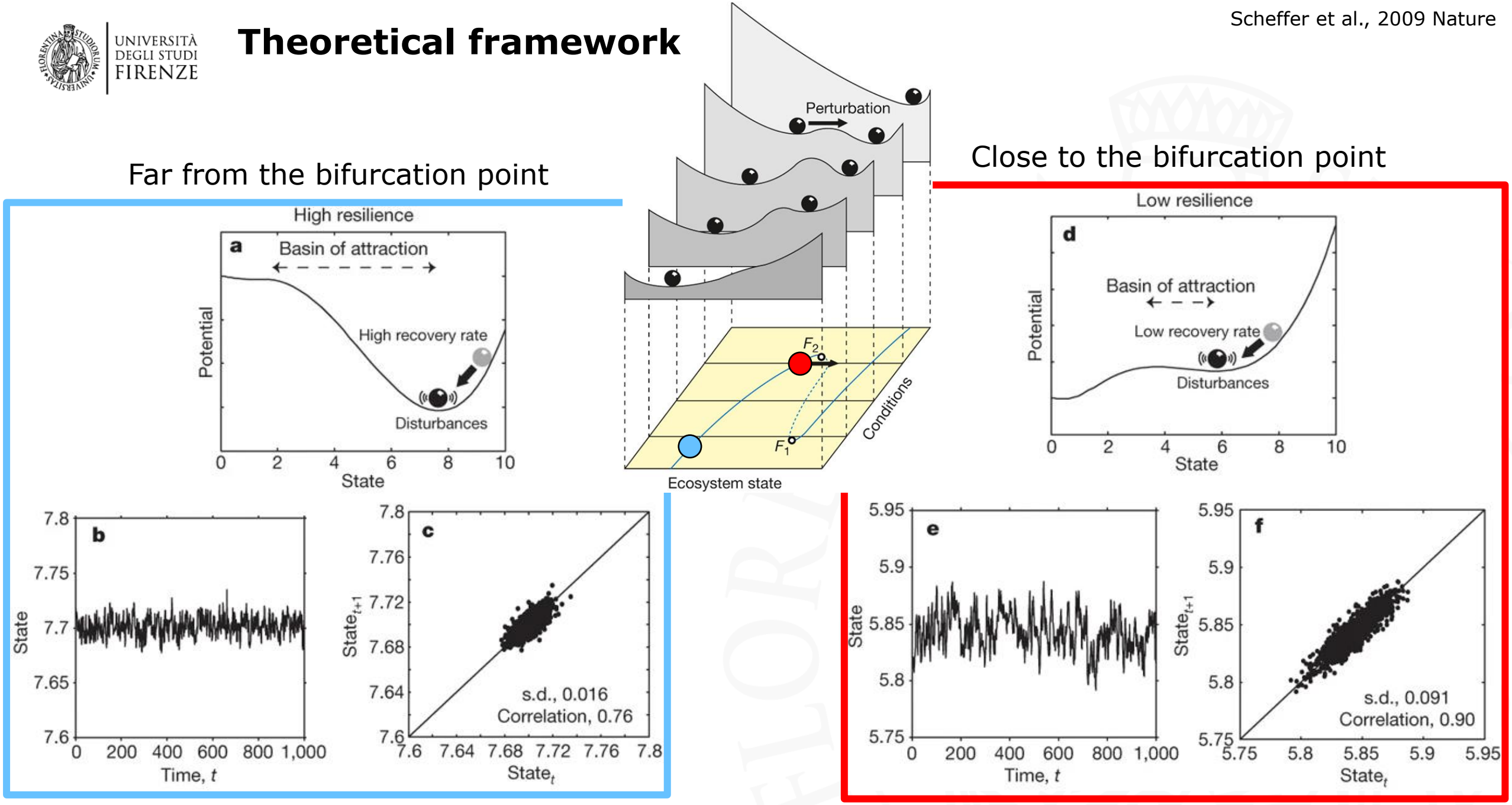


Theoretical framework

Scheffer et al., 2009 Nature

Far from the bifurcation point

Close to the bifurcation point

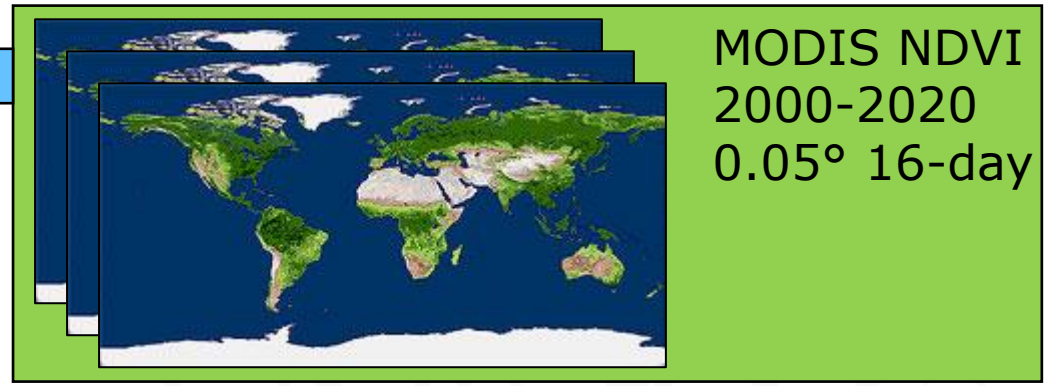


Methods

$$\mathrm{kNDVI} = \tanh\left(\mathrm{NDVI}^2\right)$$

Camps-Valls et al., 2021
Science Advances

Pre-processing
Dakos et al., 2008 PNAS



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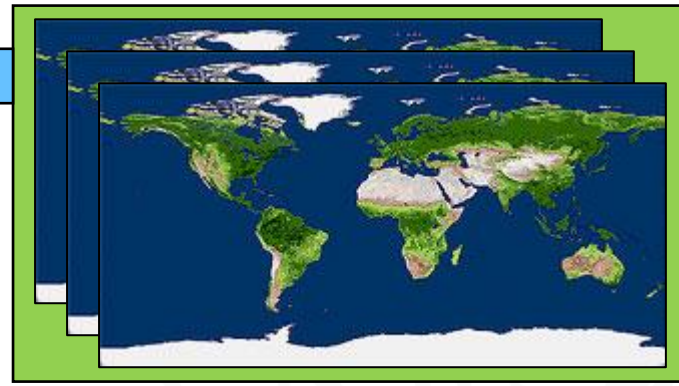
TAC^{GROSS} (CSD indicator)
Scheffer et al., 2009 Science



MODIS NDVI
2000-2020
0.05° 16-day

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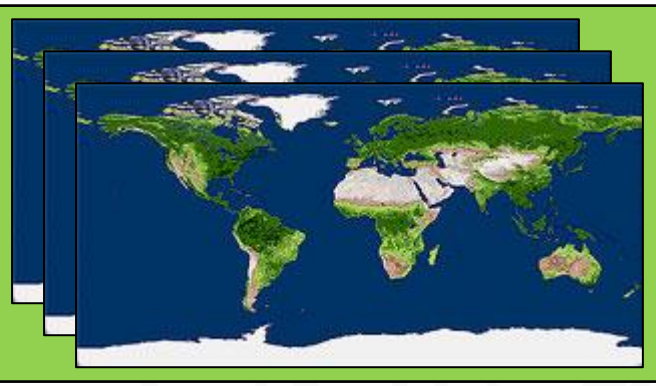

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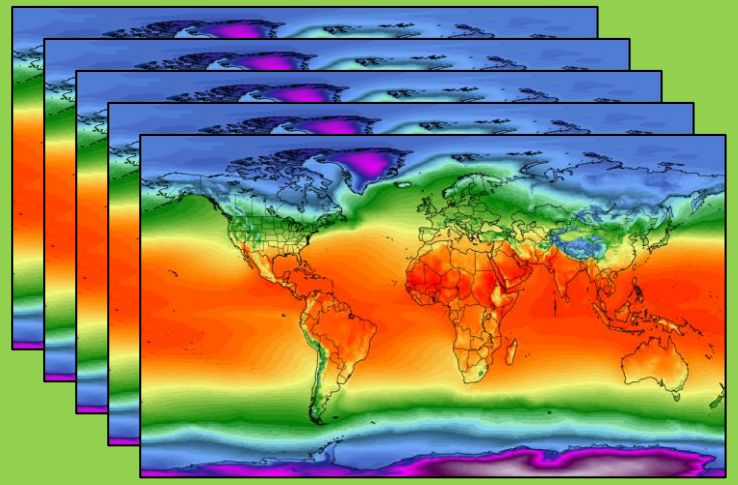

Random Forest model
$TAC^{GROSS} = f(X) + \varepsilon_f$

Environmental predictors (X)
(ESA-CCI/ERA5-Land):

- Forest density
- Background climate
- Climate variability
- Autocorrelation in climate

Climate variables: Temperature (t2m), Precipitation (tp), SW incoming radiation (ssrd), ET deficit (def)

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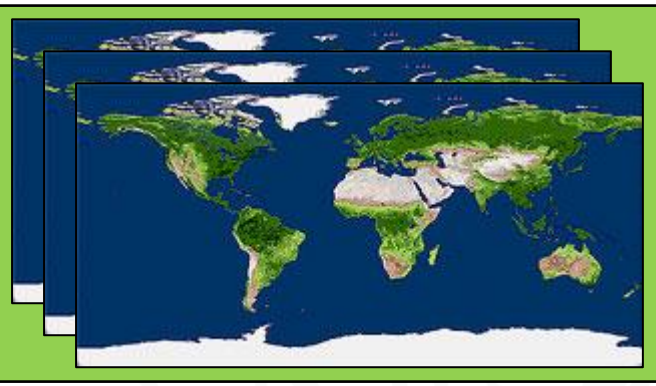
Camps-Valls et al., 2021
Science Advances



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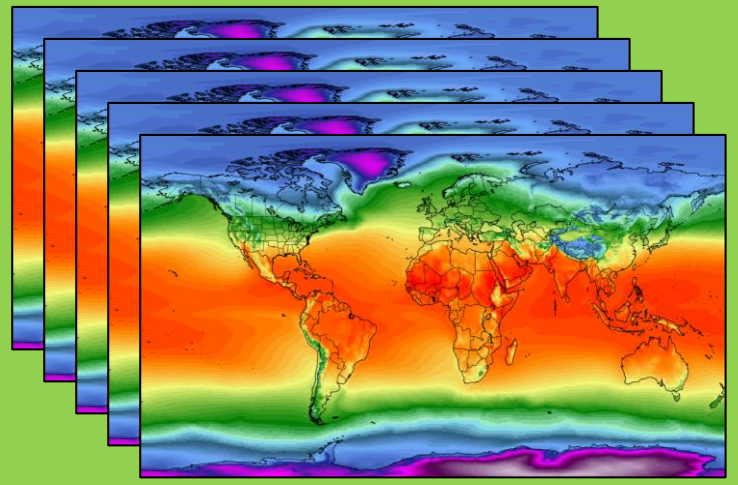
Factorial simulations
$TAC = TAC^{GROSS} - TAC^{GROSS}|X_{ac}$



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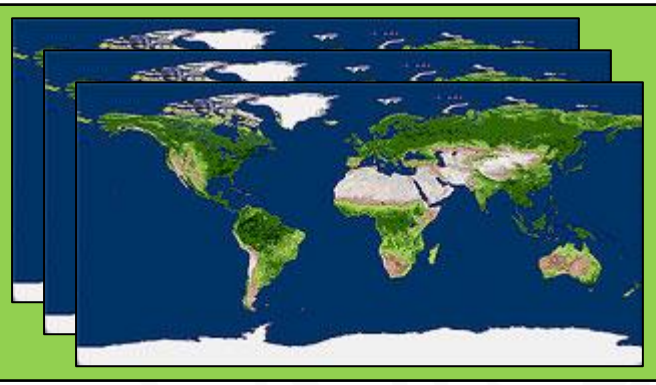

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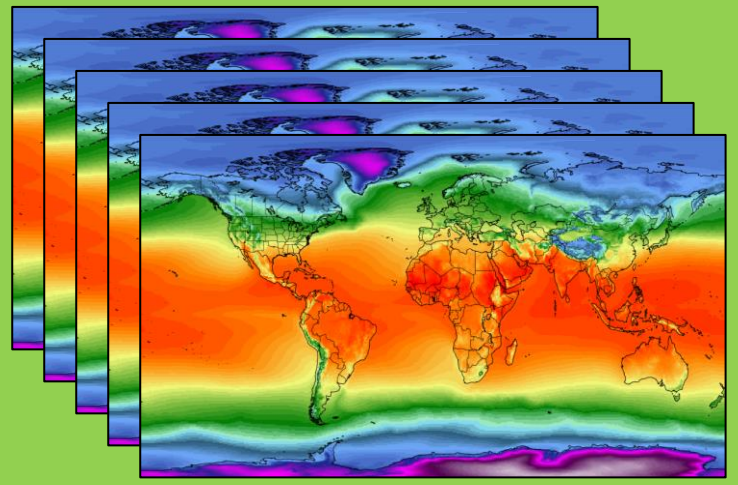

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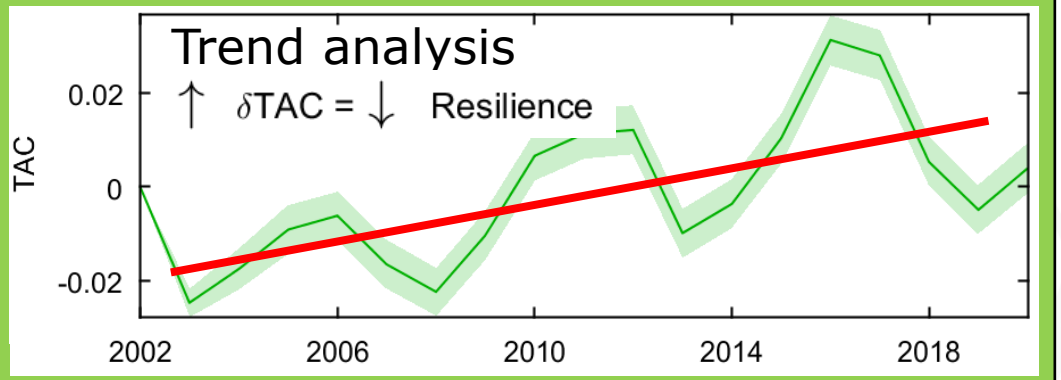
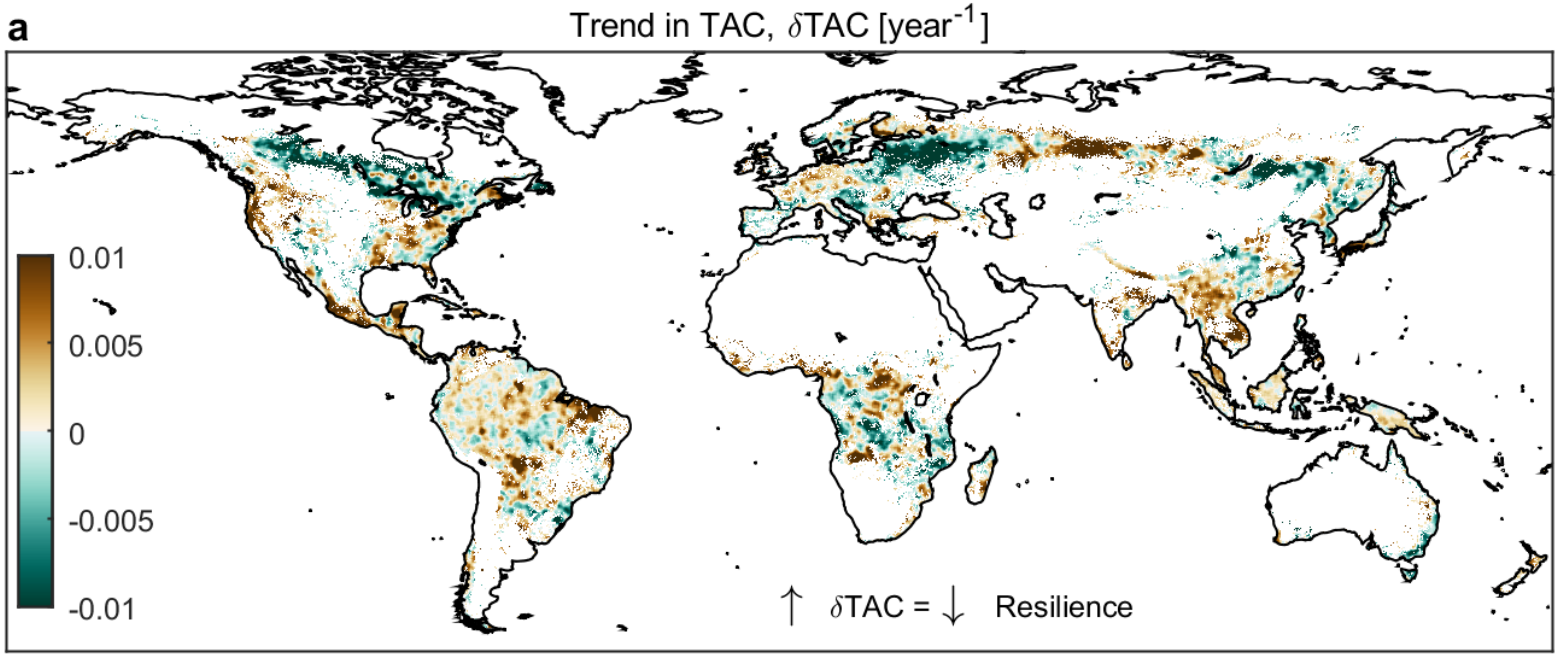
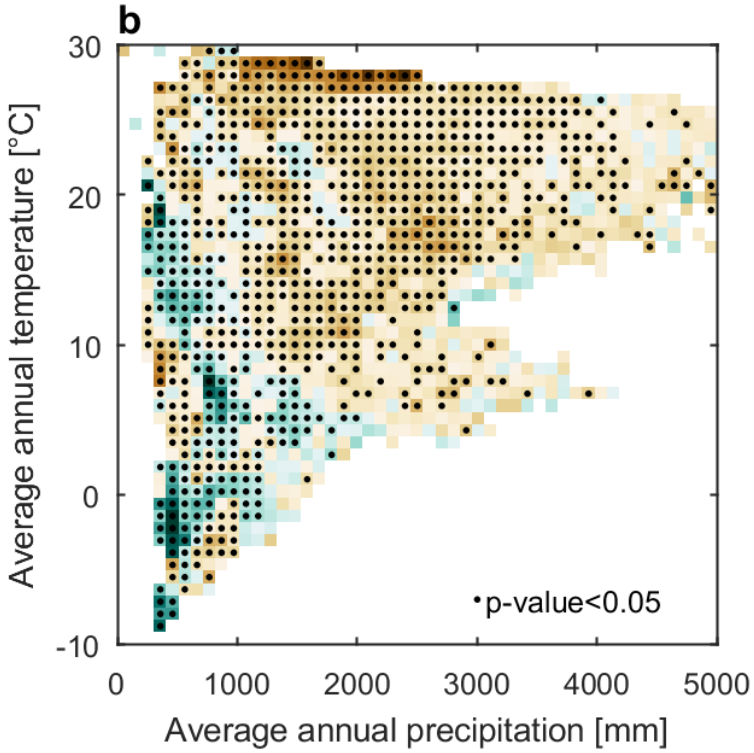
Time series of TAC forest resilience



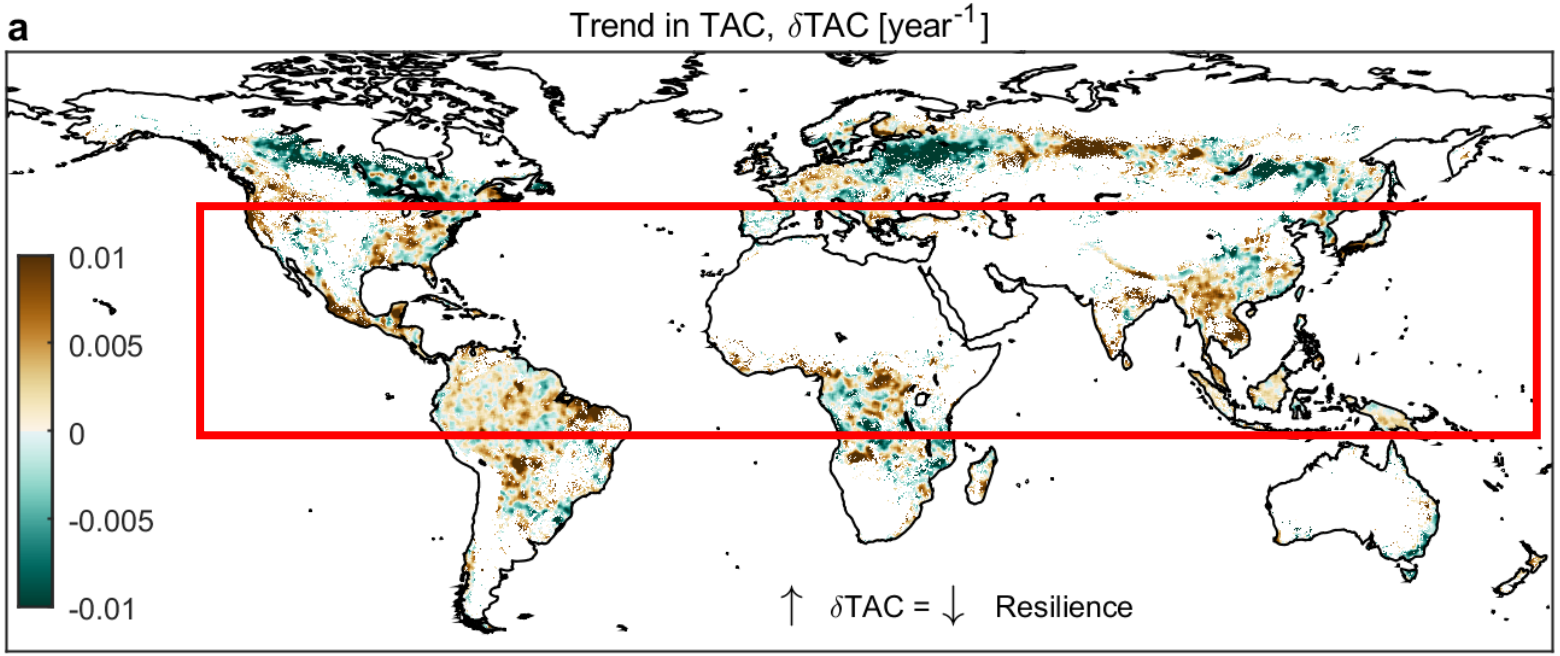
Trends and drivers of forest resilience

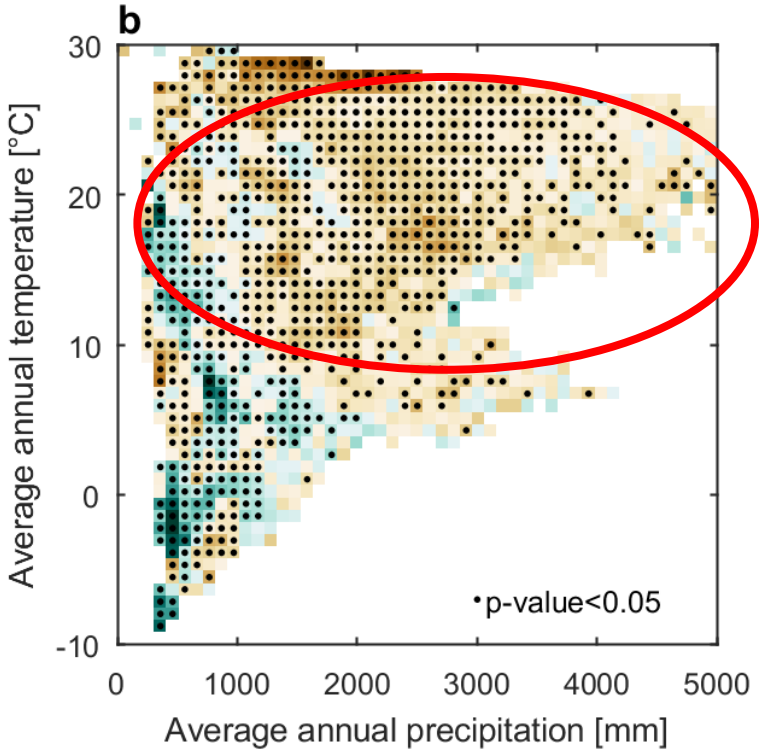




Climate regions	δTAC		
	Average	Lower ci	Upper ci
Tropical	1.63E-03	1.61E-03	1.66E-03
Arid	1.26E-03	1.19E-03	1.33E-03
Temperate	1.43E-03	1.40E-03	1.45E-03
Boreal	-1.54E-03	-1.57E-03	-1.51E-03
Globe	6.68E-04	6.53E-04	6.83E-04

Trends and drivers of forest resilience





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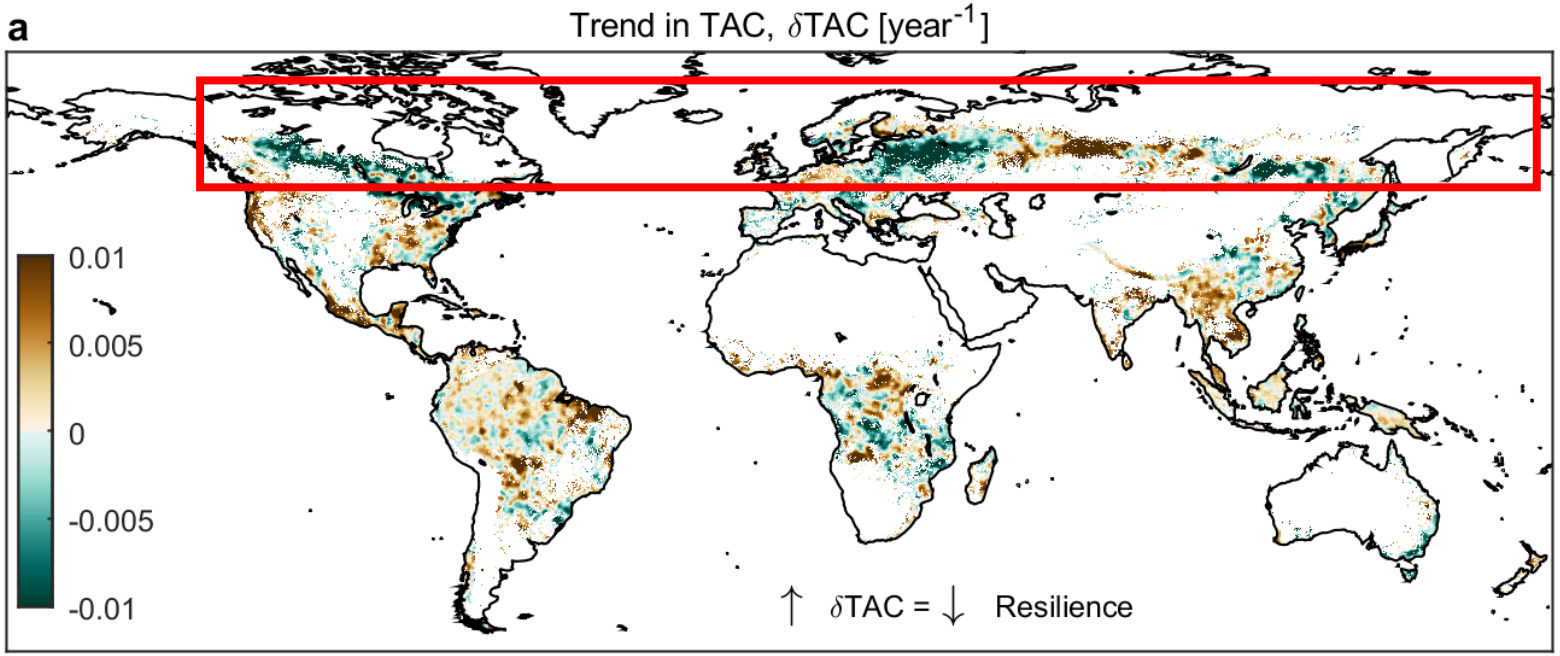
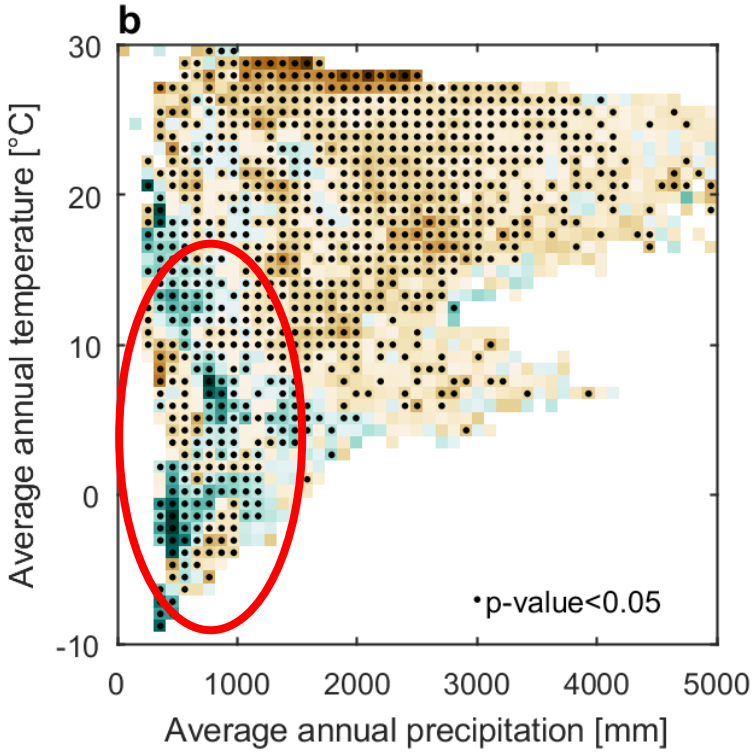
Widespread and significant increase in TAC, and thus a temporal **decline in resilience, in tropical, temperate and arid regions**.

Trends and drivers of forest resilience





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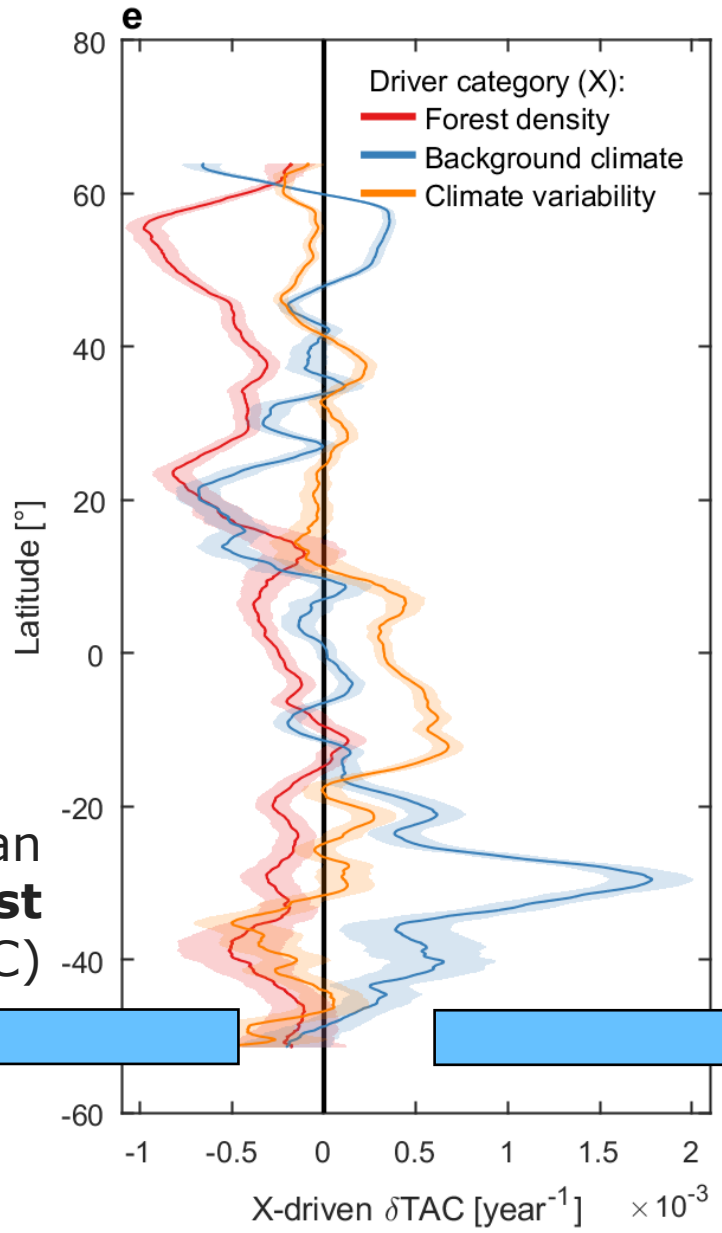
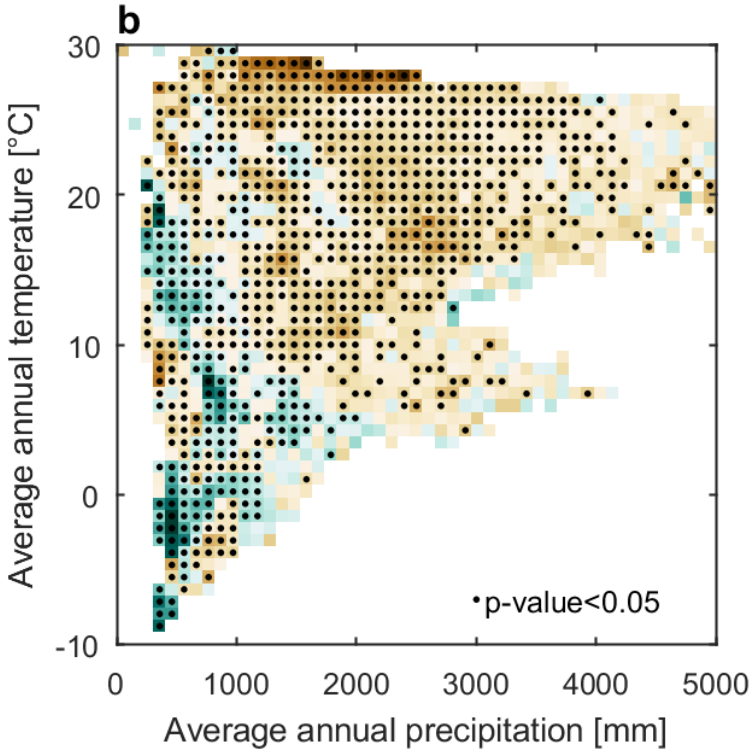
Widespread and significant increase in TAC, and thus a temporal **decline in resilience, in tropical, temperate and arid regions**.

Boreal forests show divergent local patterns with an average **increasing trend in resilience** prominently associated with a decline in TAC occurring in Eastern Canada and European Russia.

Trends and drivers of forest resilience

Variable name	Category
Forest cover	Forest density
Growing-season KNDVI	Forest density
Total precipitation	Background climate
Total precipitation	Climate variability
2-meter air temperature	Background climate
2-meter air temperature	Climate variability
Evapotranspiration deficit	Background climate
Evapotranspiration deficit	Climate variability
Surface solar radiation downwards	Background climate
Surface solar radiation downwards	Climate variability



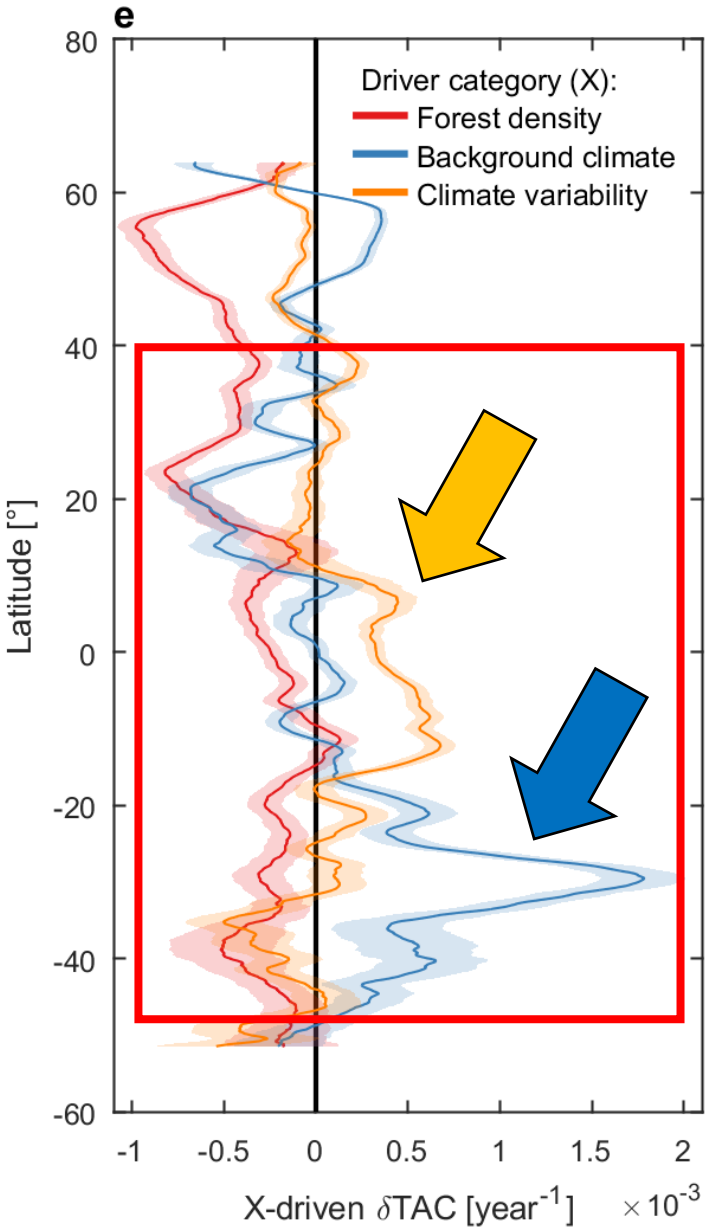


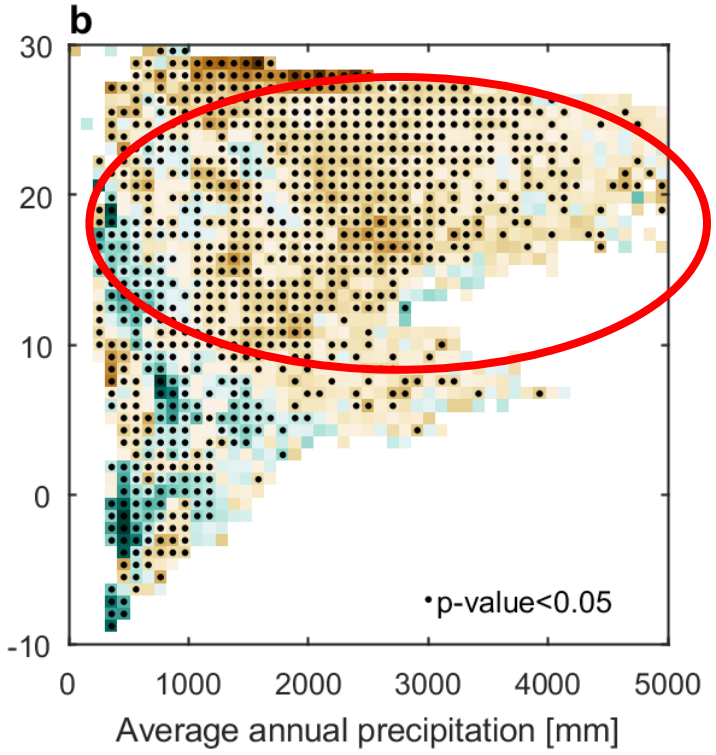
Marginal effect leading to an **increase in forest resilience** (decrease in TAC)

Marginal effect leading to a **decrease in forest resilience** (increase in TAC)

Trends and drivers of forest resilience

Tropical, arid and temperate forests are experiencing a significant decline in resilience due to **increased water limitations and climate variability**.

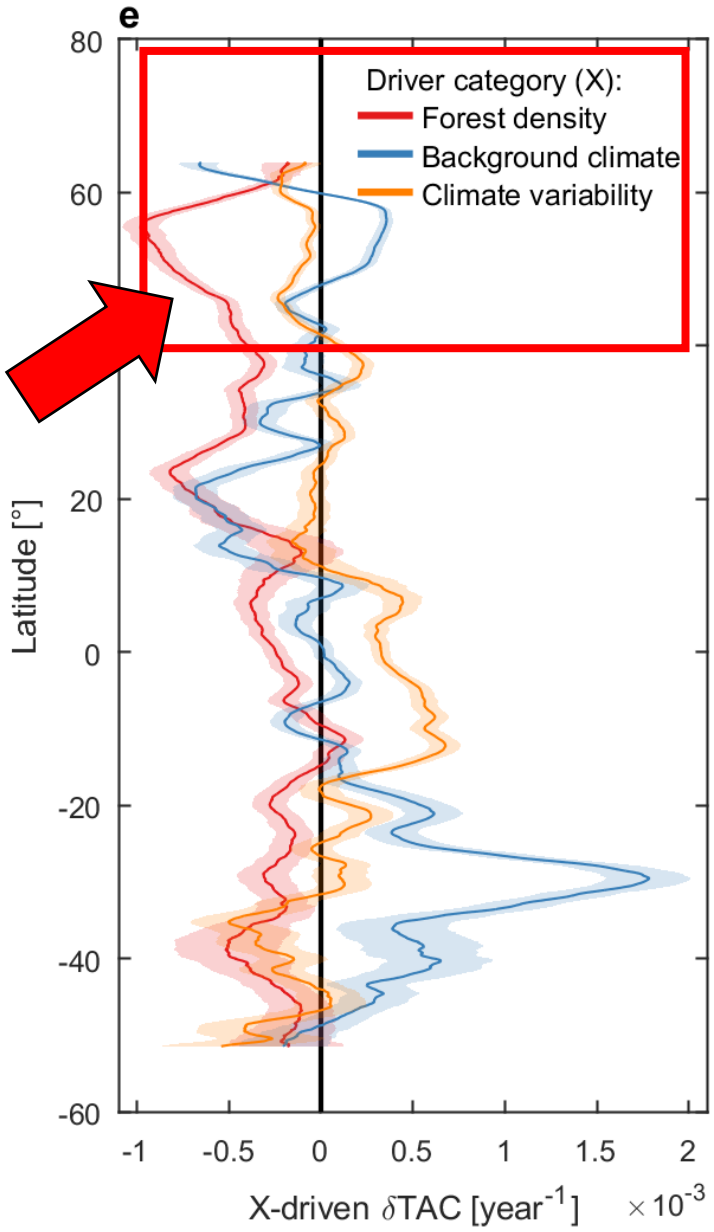


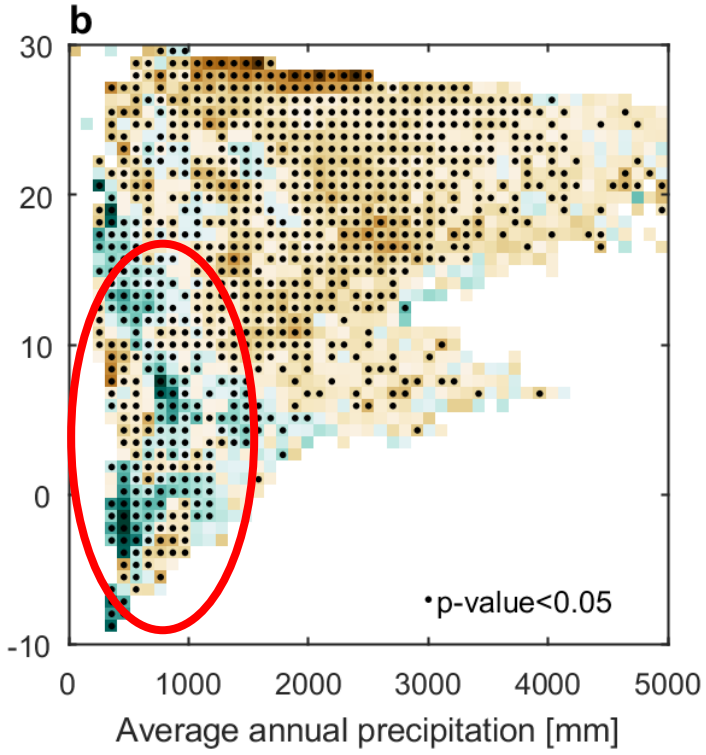


Trends and drivers of forest resilience

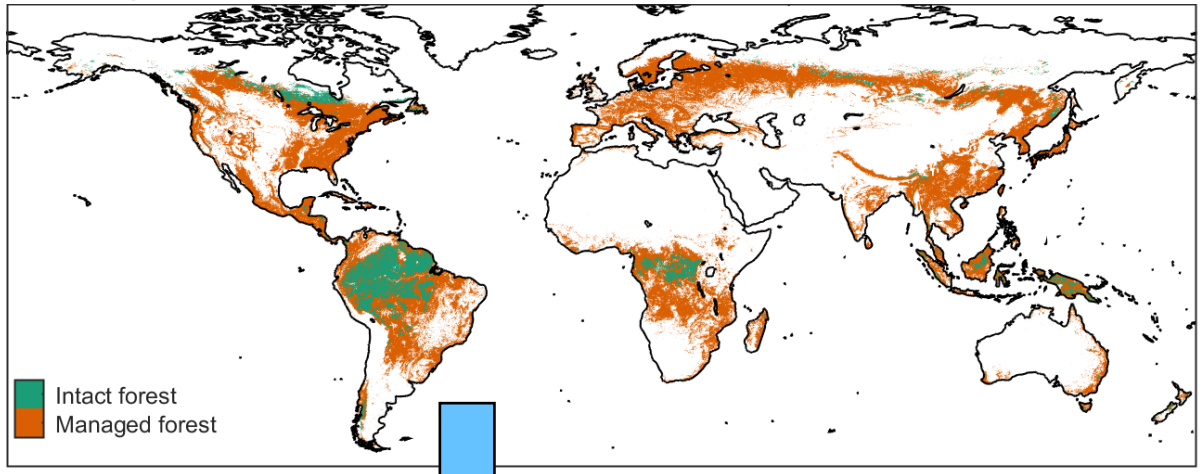
Tropical, arid and temperate forests are experiencing a significant decline in resilience due to **increased water limitations and climate variability**.

Boreal forests show an average increasing trend in resilience, probably **benefiting from warming and CO_2 fertilization**, which may outweigh the adverse effects of climate change.



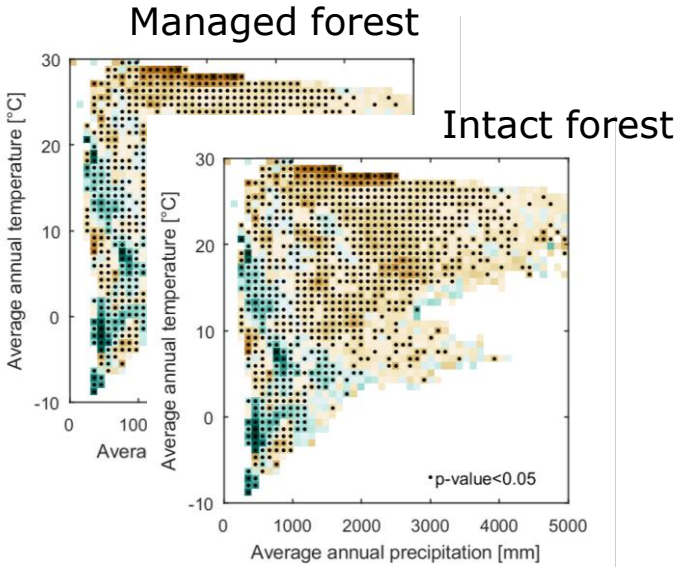


Forest management and resilience



Potapov et al., 2017
Science Advances

The resulting distributions have the same sample size, and each pair of elements shares the same background climate.



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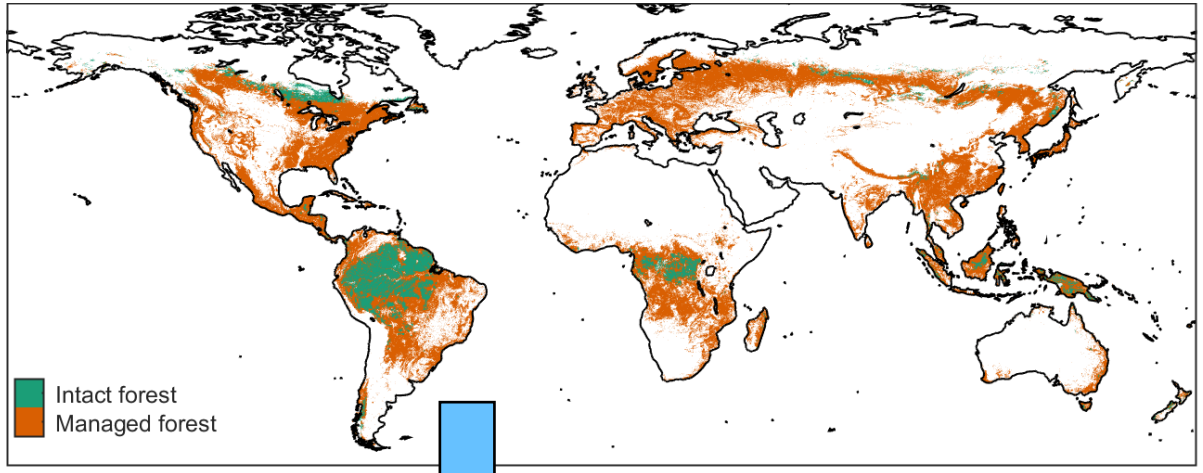
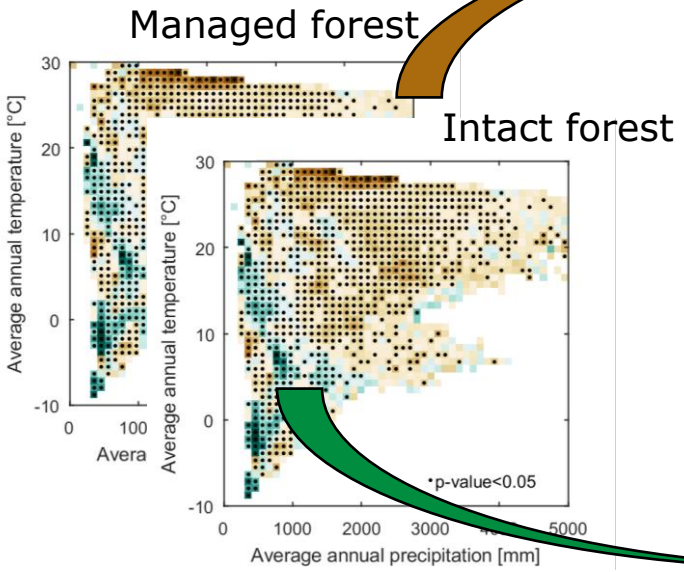


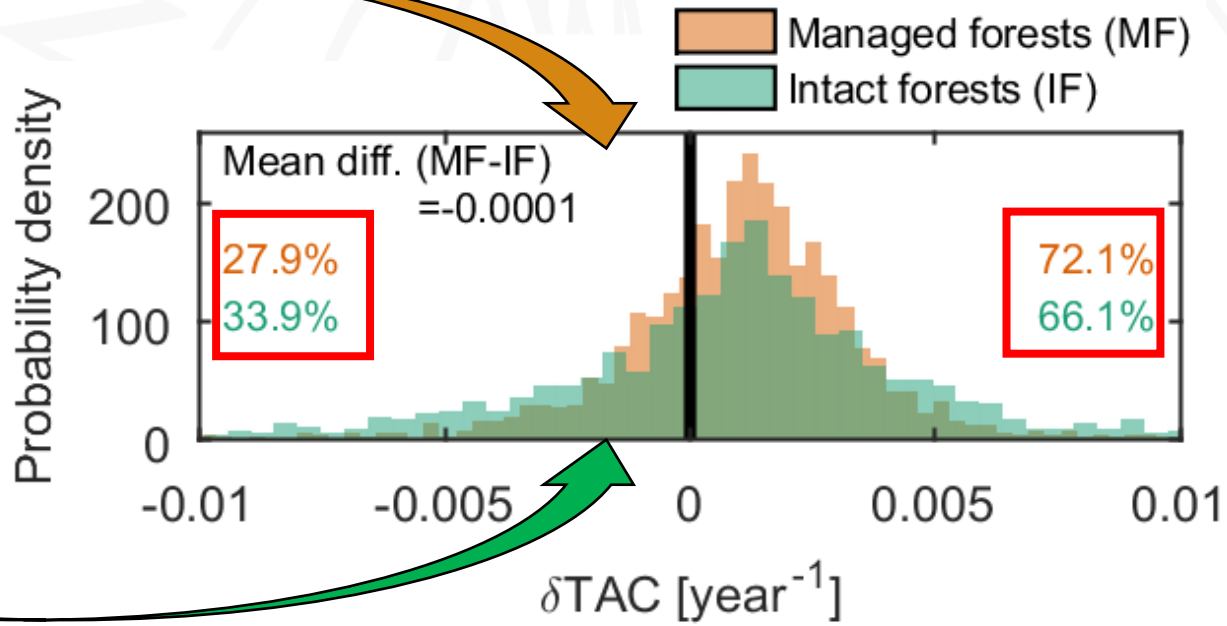
Potapov et al., 2017
Science Advances

Managed and intact forests do not present significant differences in trends of forest resilience. This corroborates the **existence of common large-scale climate drivers**.

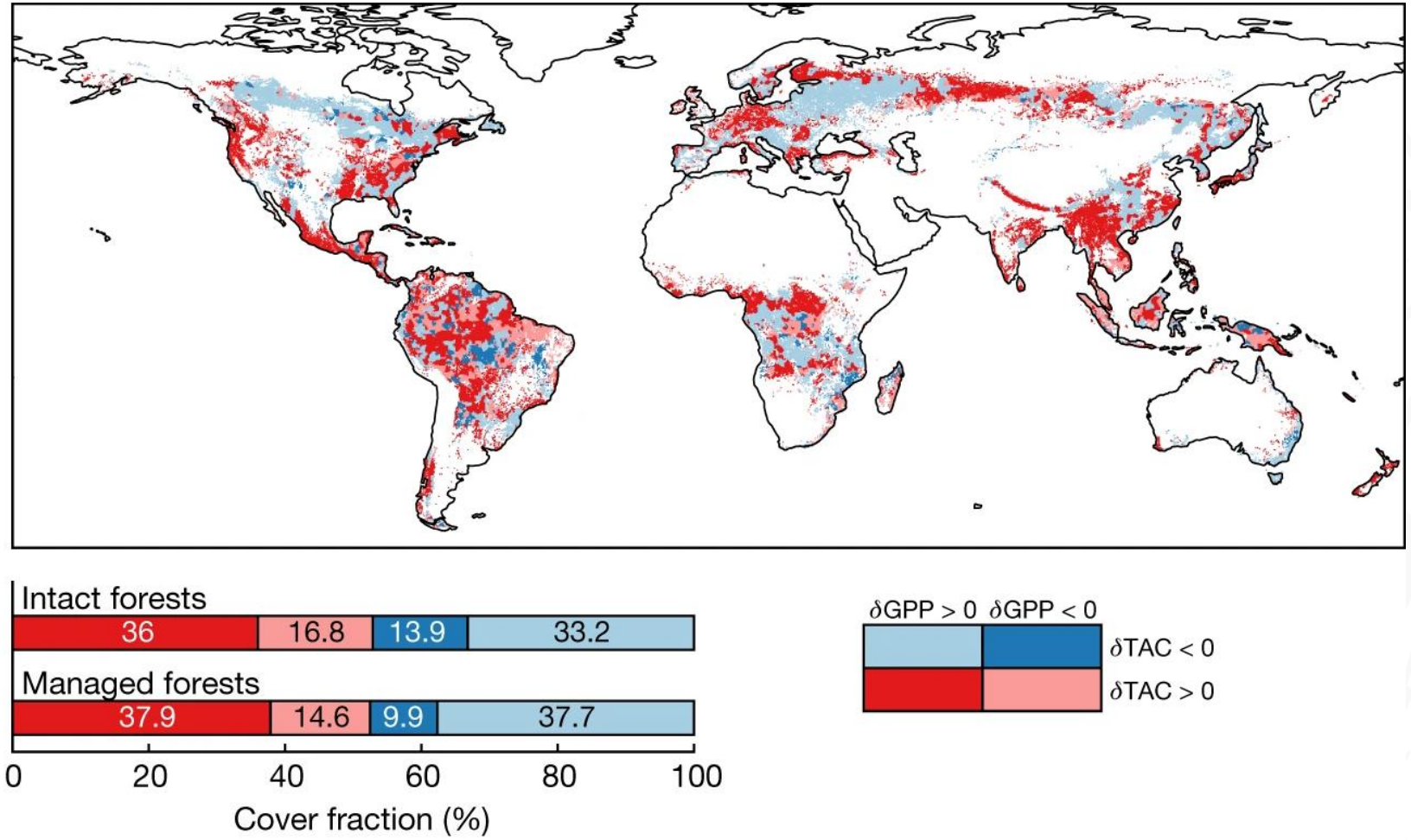
T-test to analyze the significance of the difference in the sampled means
Numbers in red boxes refer to the percentage of the observations lower and greater than zero

The resulting distributions have the same sample size, and each pair of elements shares the same climate background.

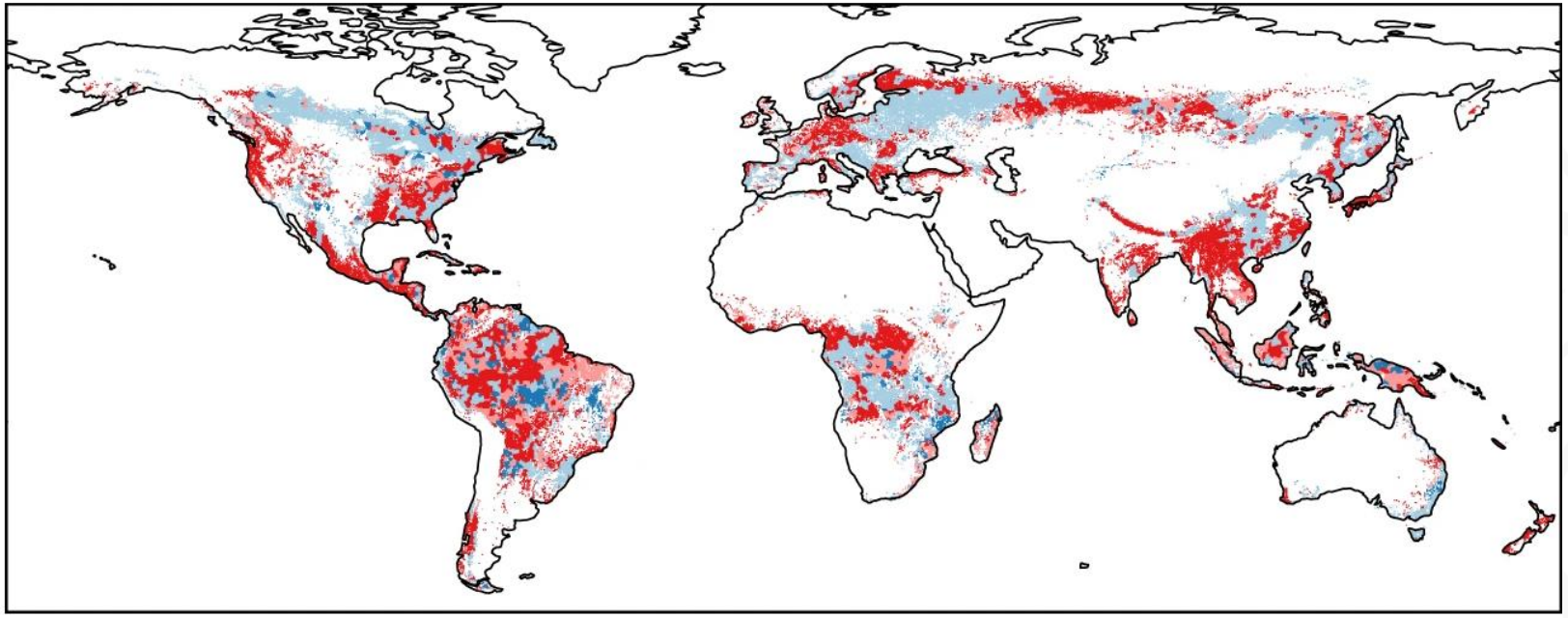


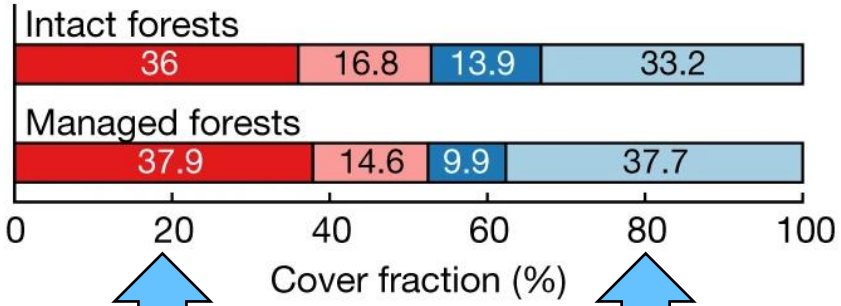


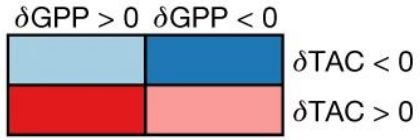
Resilience and primary productivity



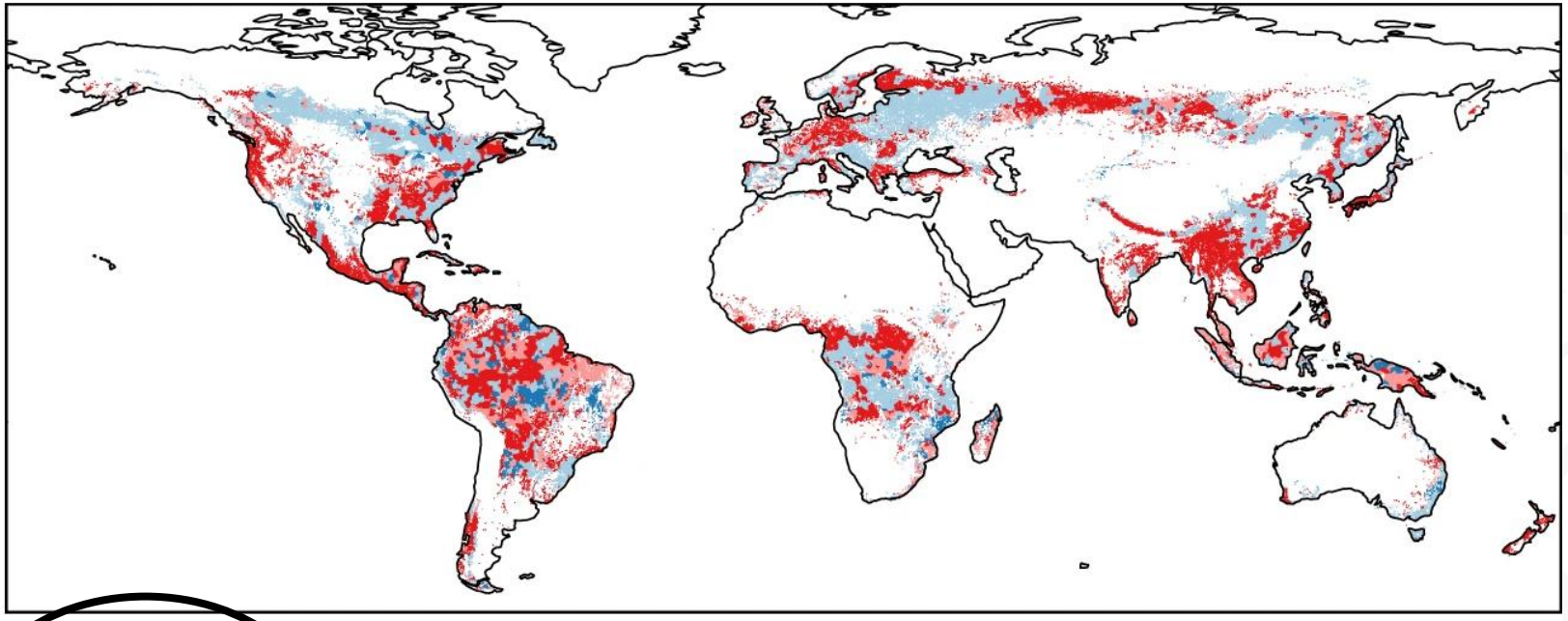
Resilience and primary productivity

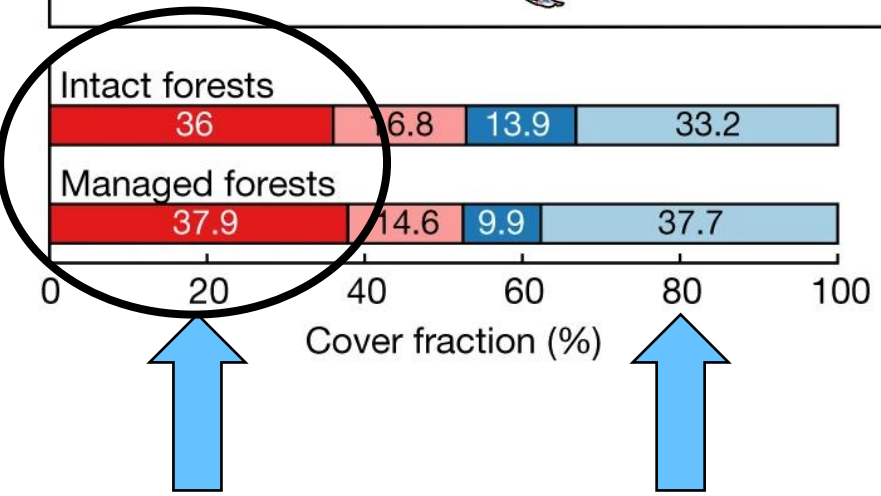


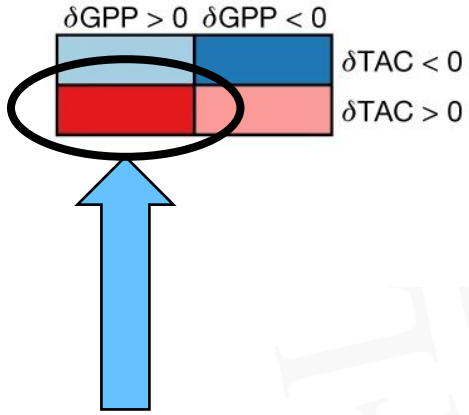




Resilience and primary productivity

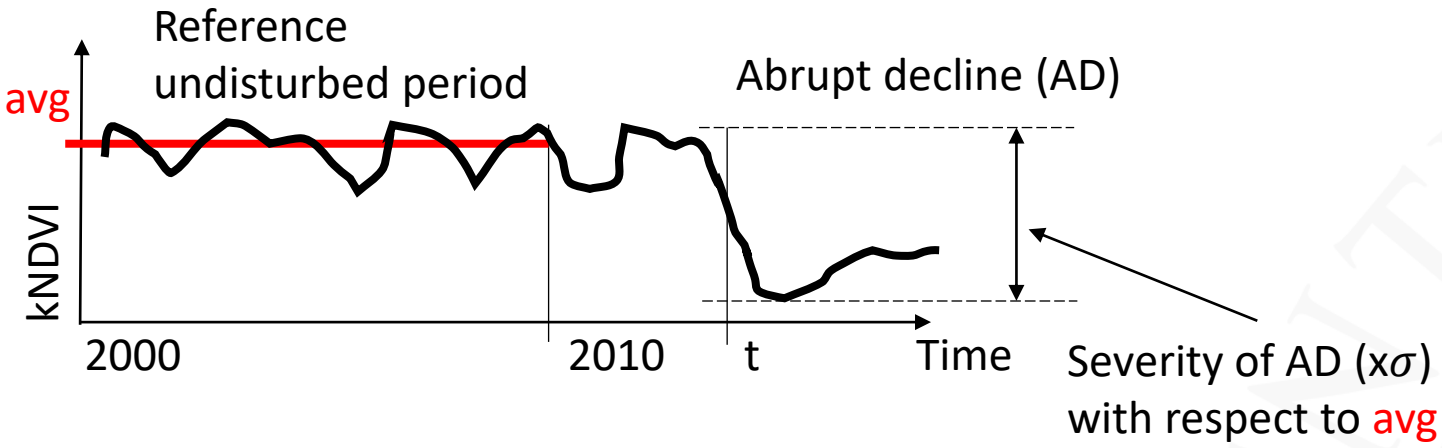






A considerable fraction of forest area is increasing primary productivity while also experiencing a declining resilience, therefore leading to an expanding but more vulnerable forest sink

Early signals of abrupt forest decline



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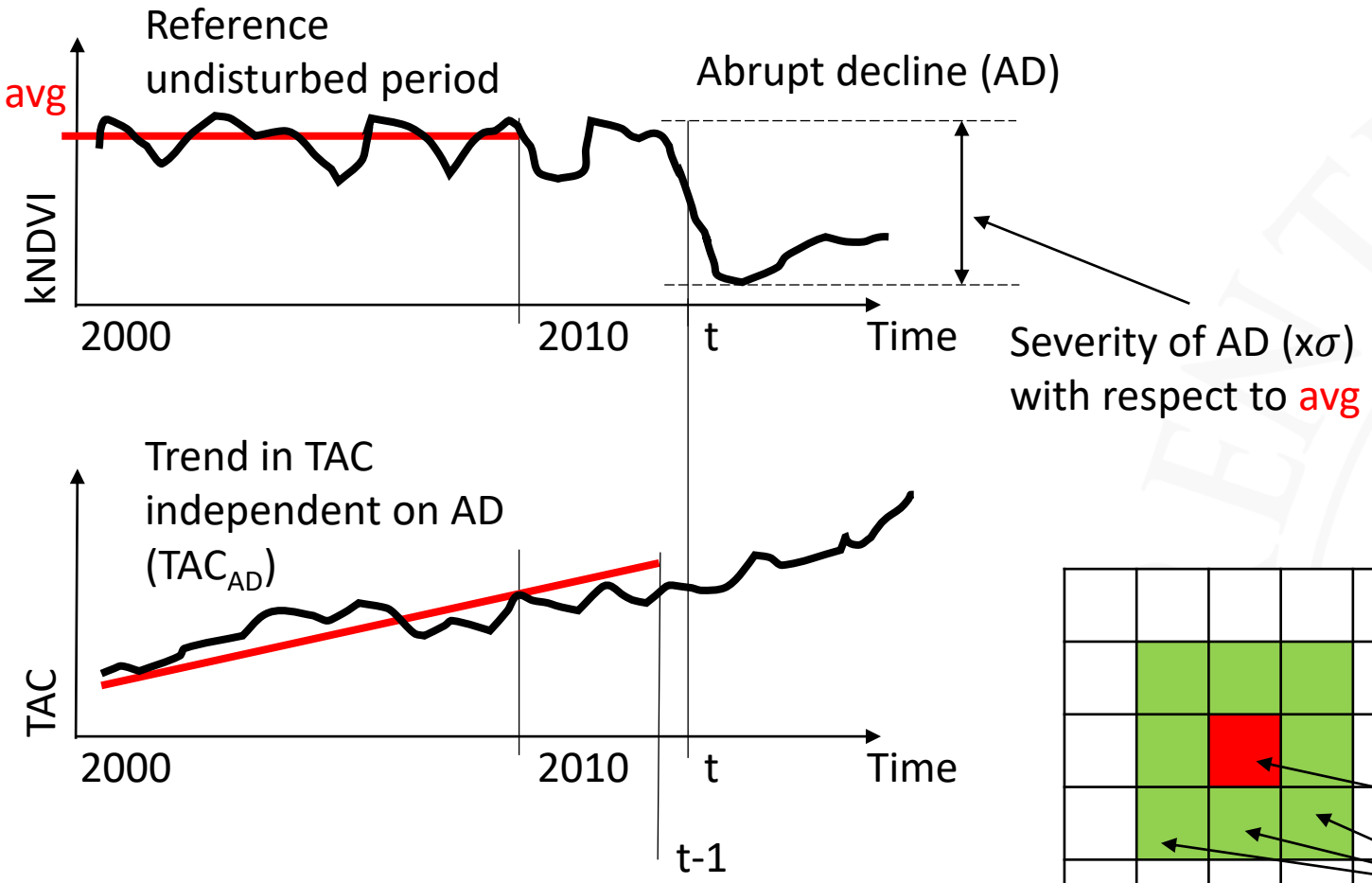
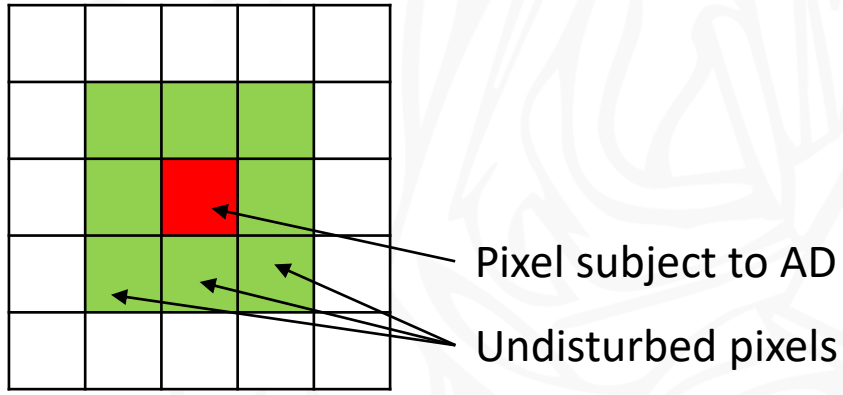
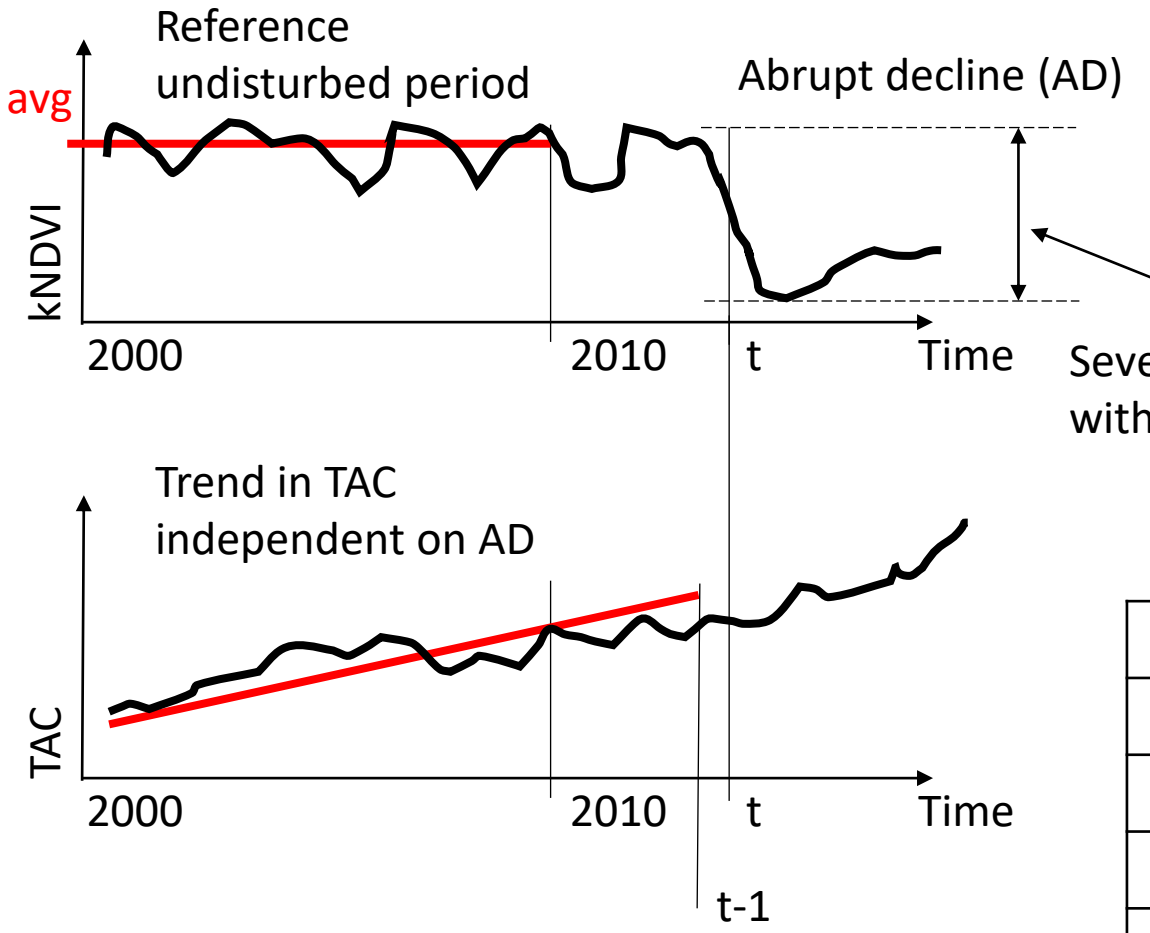
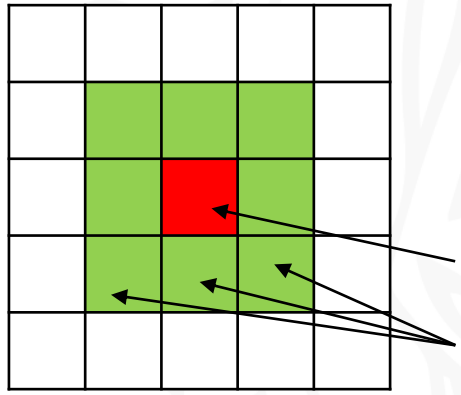


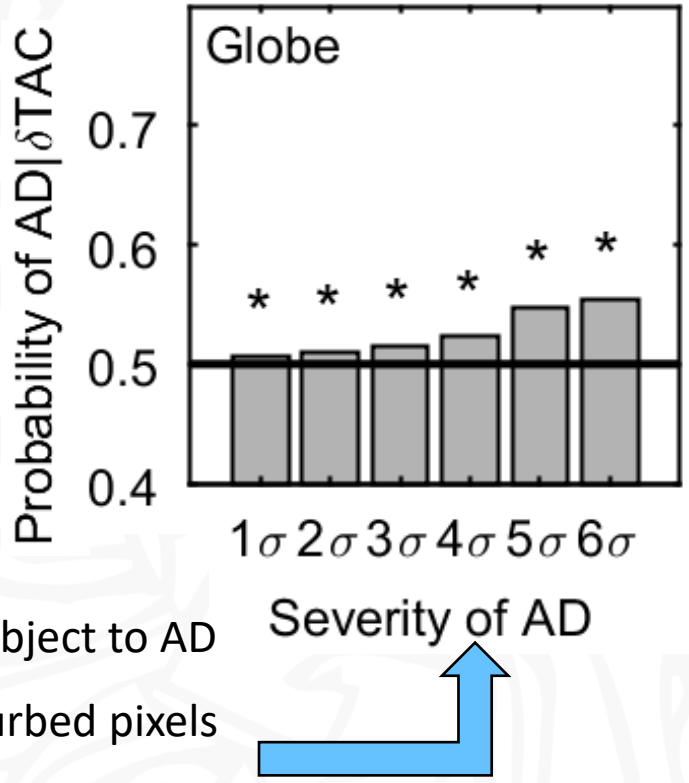


Early signals of abrupt forest decline

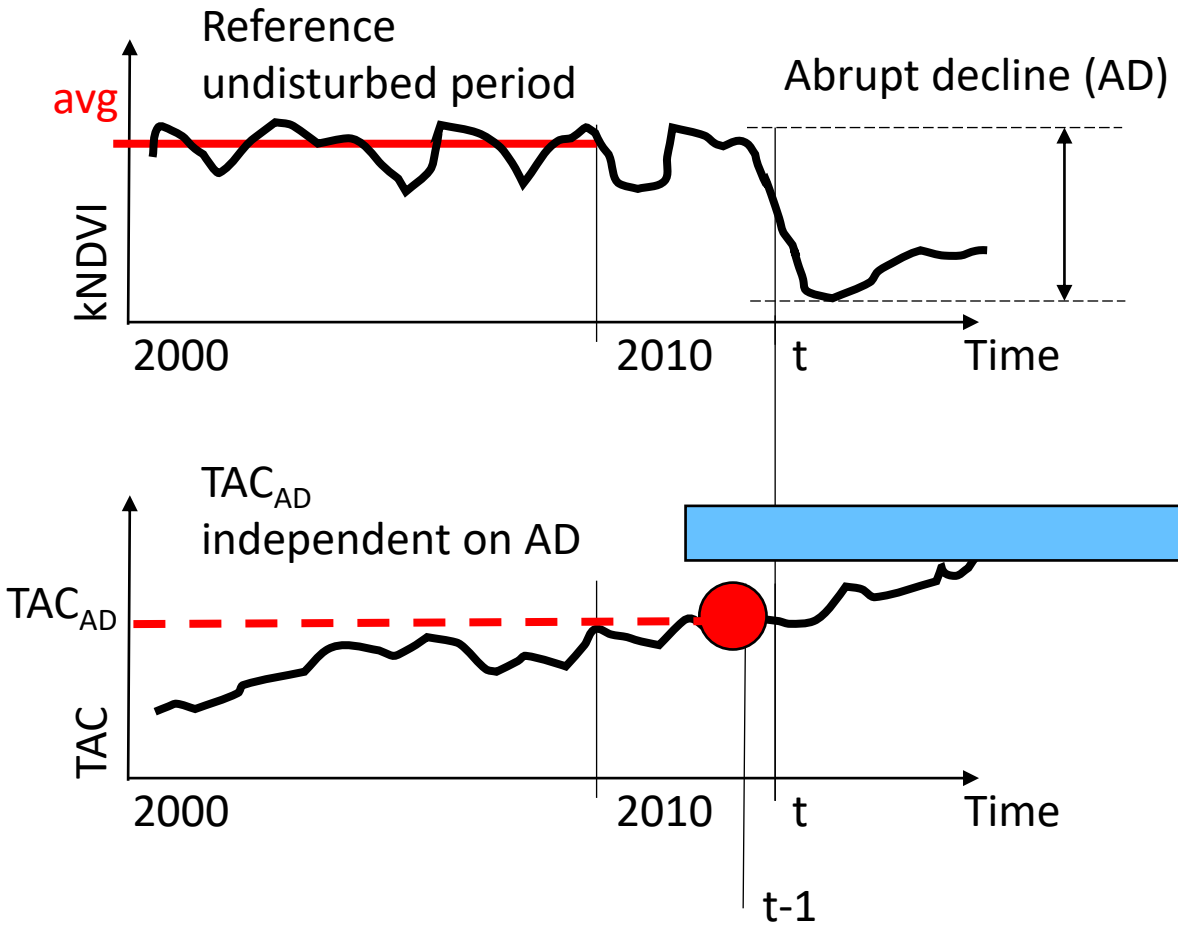
Global intact forests have a probability of AD conditional on δTAC greater than 0.5. This signal is statistically significant and increases with the severity of AD, suggesting that **the progressive deterioration of ecosystem states has probably contributed to the upsurge of negative anomalies** in forest dynamics.



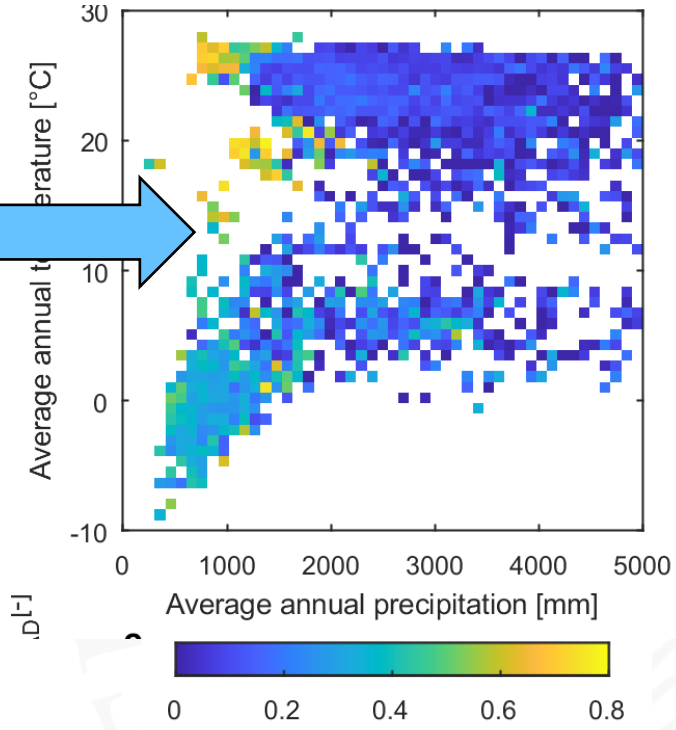




Critical threshold mechanisms

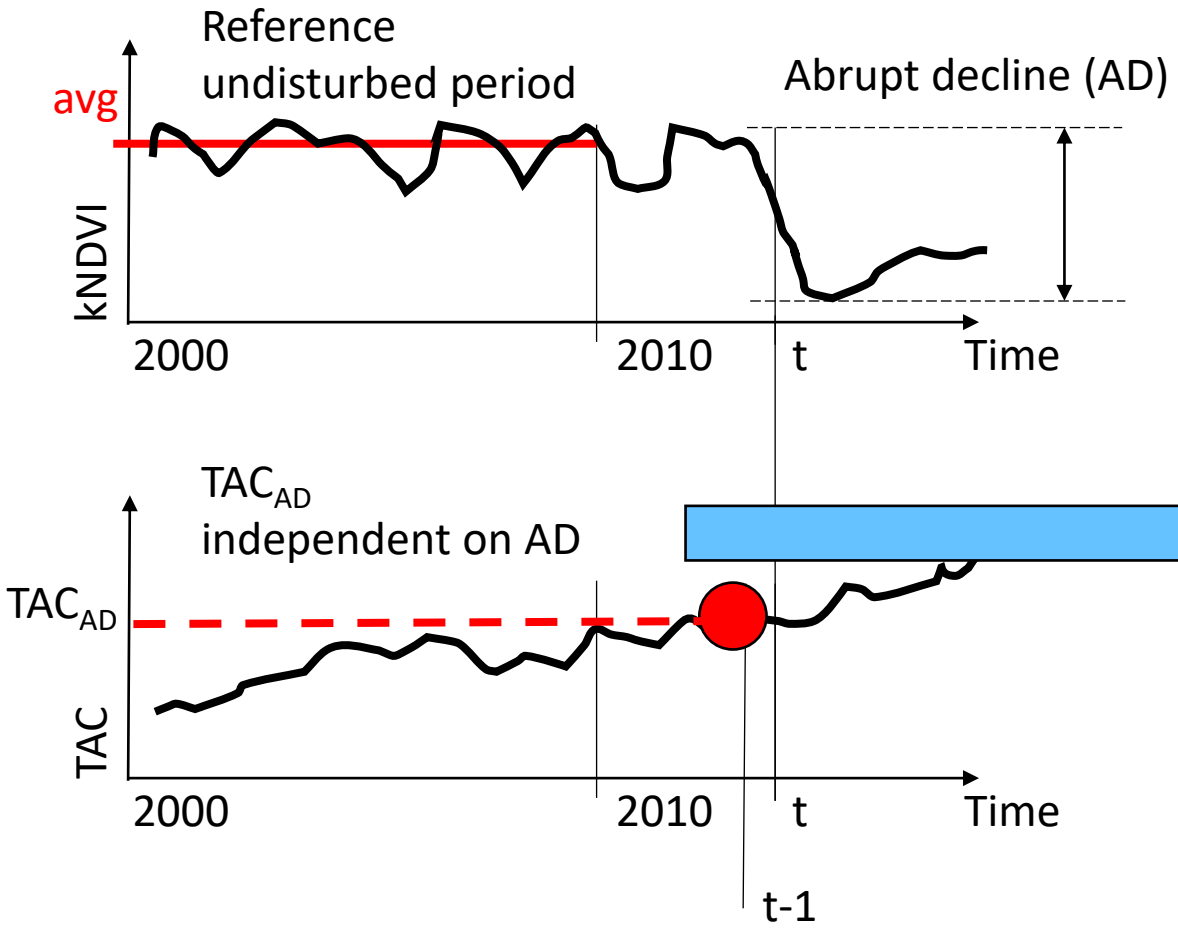


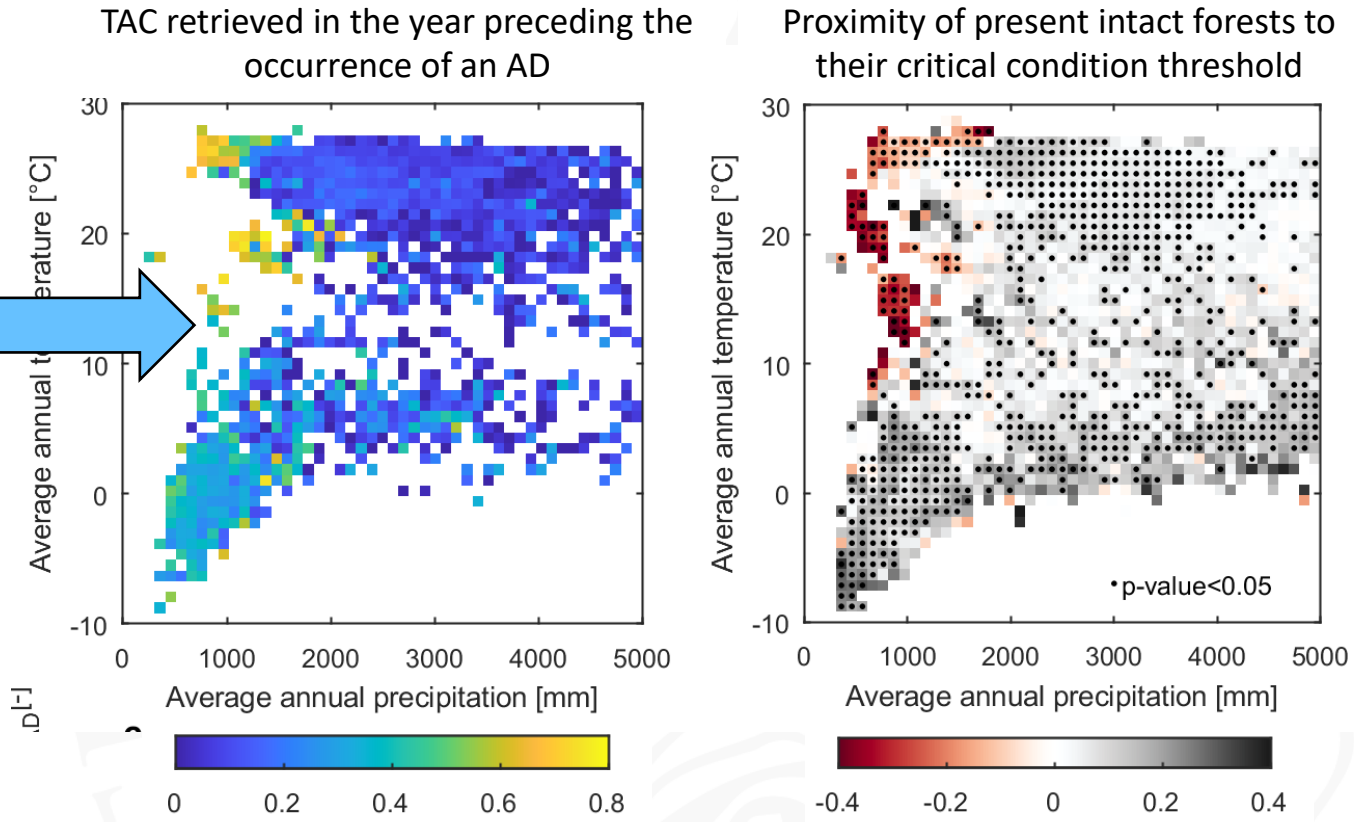
TAC retrieved in the year preceding the occurrence of an AD



Critical threshold mechanisms

In 2020, about **23% of intact forests (3.32 Pg C of GPP) have reached their TAC_{AD}** and are experiencing a concomitant increase in δTAC, therefore implying an ongoing reduction in resilience to levels already close to an AD and, potentially, to a tipping point





Conclusions

Together, these signals reveal a **widespread decline in the capacity of forests to withstand perturbations** that should be accounted for in the design of **land-based mitigation and adaptation plans**.

Global data-driven assessment shows that resilience thinking can be developed effectively in a science-based and solution-oriented framework **to support the many challenges of forest management** in times of rapid climatic changes.

Conclusions

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Global data-driven assessment shows that resilience thinking can be developed effectively in a science-based and solution-oriented framework **to support the many challenges of forest management** in times of rapid climatic changes.

Article

Emerging signals of declining forest resilience under climate change

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Giovanni Forzieri[1,5], Vasilis Dakos[2], Nate G. McDowell[3,4], Alkama Ramdane[1] & Alessandro Cescatti[1]

Forest ecosystems depend on their capacity to withstand and recover from natural and anthropogenic perturbations (that is, their resilience)[1]. Experimental evidence of sudden increases in tree mortality is raising concerns about variation in forest resilience[2], yet little is known about how it is evolving in response to climate change. Here we integrate satellite-based vegetation indices with machine learning to show how forest resilience, quantified in terms of critical slowing down indicators[3–5], has changed during the period 2000–2020. We show that tropical, arid and temperate forests are experiencing a significant decline in resilience, probably related to increased water limitations and climate variability. By contrast, boreal forests show divergent local patterns with an average increasing trend in resilience, probably benefiting from warming and CO_2 fertilization, which may outweigh the adverse effects of climate change. These patterns emerge consistently in both managed and intact forests, corroborating the existence of common large-scale climate drivers. Reductions in resilience are statistically linked to abrupt declines in forest primary productivity, occurring in response to slow drifting towards a critical resilience threshold. Approximately 23% of intact undisturbed forests, corresponding to 3.32 Pg C of gross primary productivity, have already reached a critical threshold and are experiencing a further degradation in resilience. Together, these signals reveal a widespread decline in the capacity of forests to withstand perturbation that should be accounted for in the design of land-based mitigation and adaptation plans.

Forzieri et al., 2022 Nature

UNIVERSITÀ DEGLI STUDI FIRENZE

Postdoctoral research position at UNIFI within the ClimTip Horizon project (application opening in April 2024)

Climate-related tipping points - Ecosystem adaptation strategies and implications for mitigation

Thank you

Emerging signals of a global drift in forest resilience under climate change

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